

- Homework: 5.39, 5.48 in the attachment
- Tutorial: Second homework problem of Chapter 7B, 8.44, 8.46

- 5.39 Twenty-four voice signals are sampled uniformly and are then time-division multiplexed, using PAM. The PAM signal is reconstructed from flat-topped pulses with $1\text{-}\mu\text{s}$ duration. The multiplexing operation provides for synchronization by adding an extra pulse of sufficient amplitude and also $1\text{-}\mu\text{s}$ duration. The highest frequency component of each voice signal is 3.4 kHz .
- Assuming a sampling rate of 8 kHz , calculate the spacing between successive pulses of the multiplexed signal.
 - Repeat your calculation, assuming the use of Nyquist rate sampling.

- 5.48 Consider a multiplex system in which four input signals $m_1(t)$, $m_2(t)$, $m_3(t)$, and $m_4(t)$ are respectively multiplied by the carrier waves

$$\begin{aligned} & [\cos(\omega_a t) + \cos(\omega_b t)], \\ & [\cos(\omega_a t + \alpha_1) + \cos(\omega_b t + \beta_1)], \\ & [\cos(\omega_a t + \alpha_2) + \cos(\omega_b t + \beta_2)], \end{aligned}$$

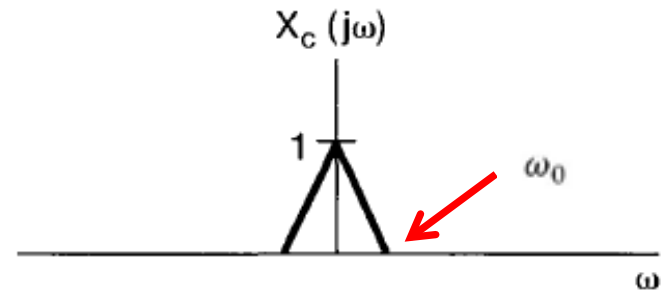
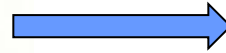
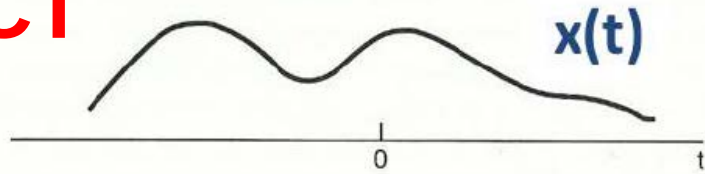
and

$$[\cos(\omega_a t + \alpha_3) + \cos(\omega_b t + \beta_3)] \quad \text{S \& S Solution}$$

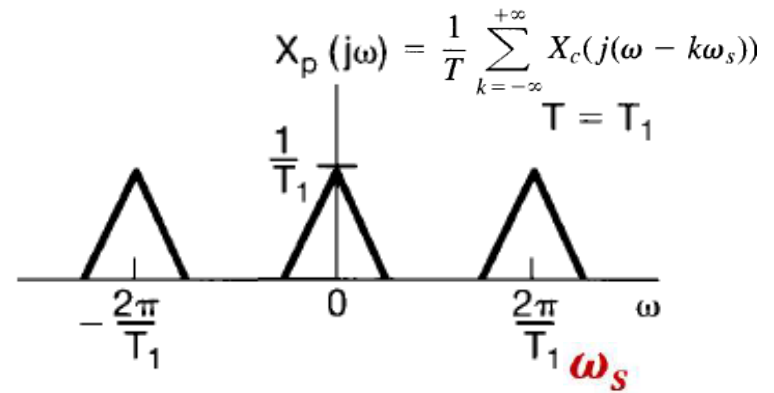
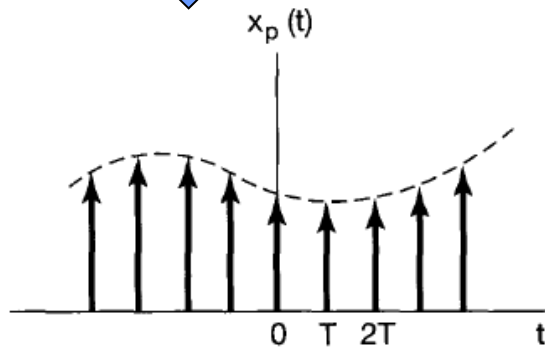
and the resulting DSB-SC signals are summed and then transmitted over a common channel. In the receiver, demodulation is achieved by multiplying the sum of the DSB-SC signals by the four carrier waves separately and then using filtering to remove the unwanted components. Determine the conditions that the phase angles $\alpha_1, \alpha_2, \alpha_3$ and $\beta_1, \beta_2, \beta_3$ must satisfy in order that the output of the k th demodulator be $m_k(t)$, where $k = 1, 2, 3, 4$.

Review

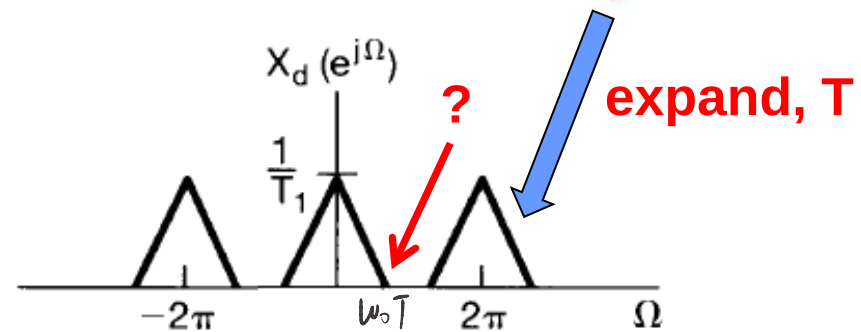
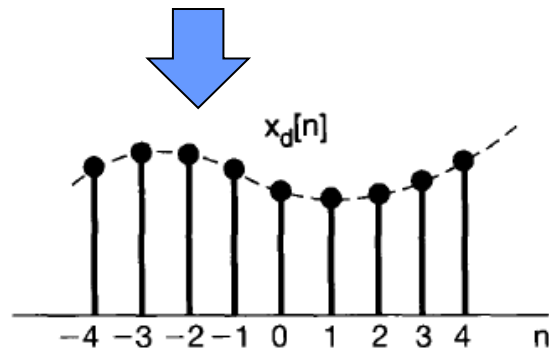
CT



CT



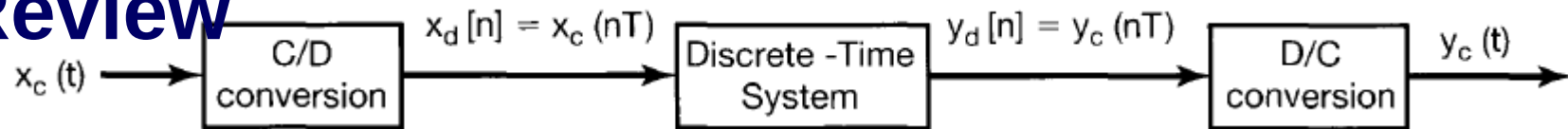
DT



$$\Omega = \omega T$$

$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

Review



$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$X_d(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

$$Y(j\omega) = \left[\frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)) \right] H_D(e^{j\omega T}) H_{LP}(j\omega)$$

$$= X_c(j\omega) \tilde{H}_D(e^{j\omega T}) \quad (1)$$

where

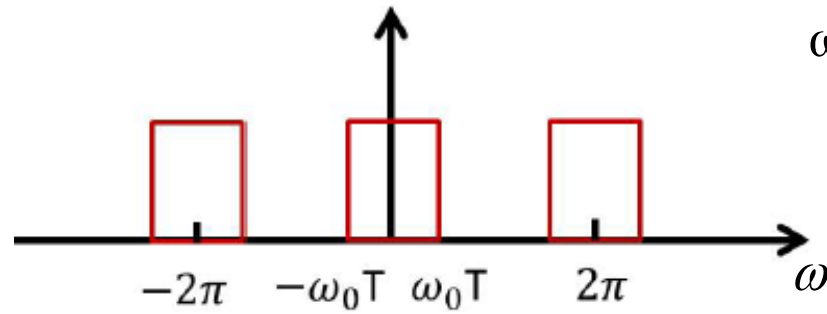
$$\tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

- It is equivalent to a continuous-time LTI system $H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$
- $H_D(e^{j\omega T})$ is a periodic extension of $\tilde{H}_D(e^{j\omega T})$ with period $\omega_s = 2\pi/T$

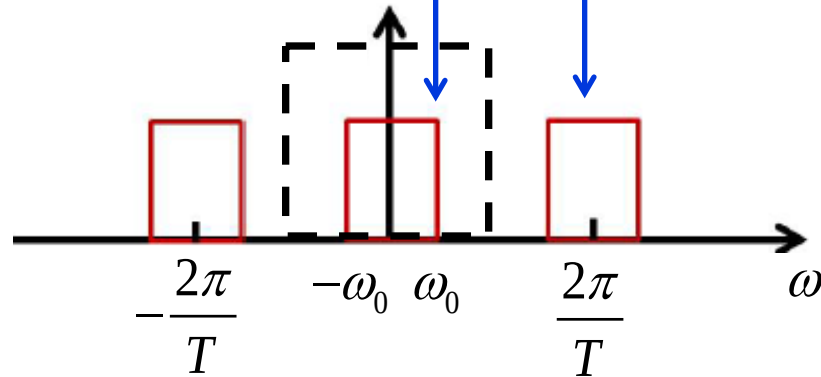
Review

T : sampling period
 $\omega_s = 2\pi/T$: sampling rate

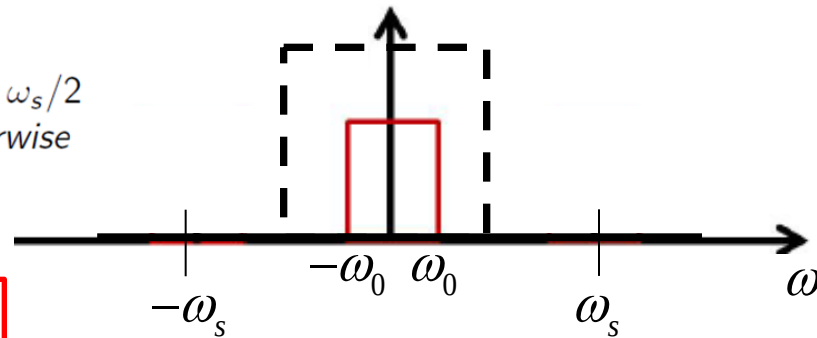
$$H_D(e^{j\omega})$$



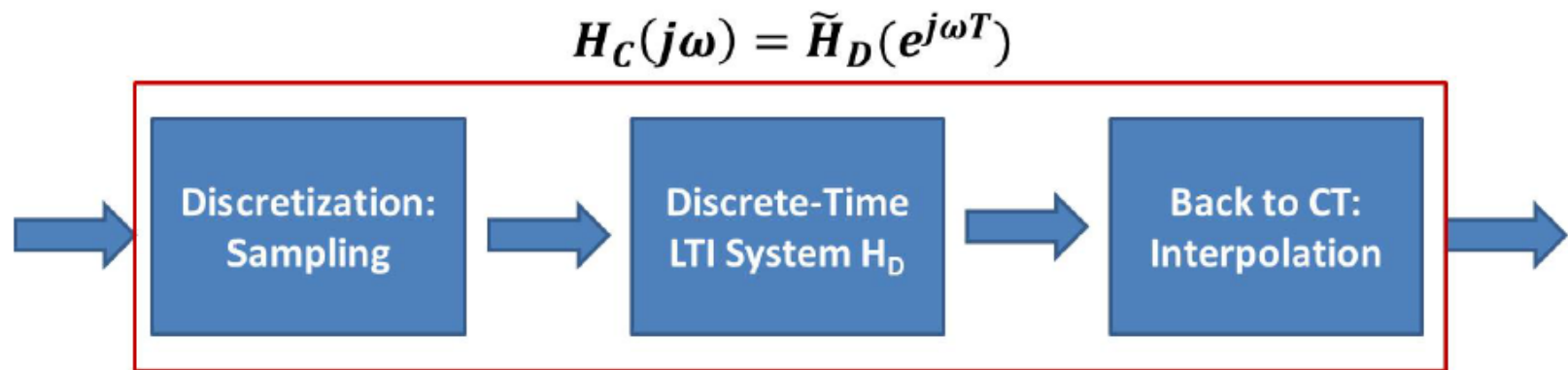
$$H_D(e^{j\omega T})$$



$$\tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & \text{otherwise} \end{cases}$$



$$H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$$

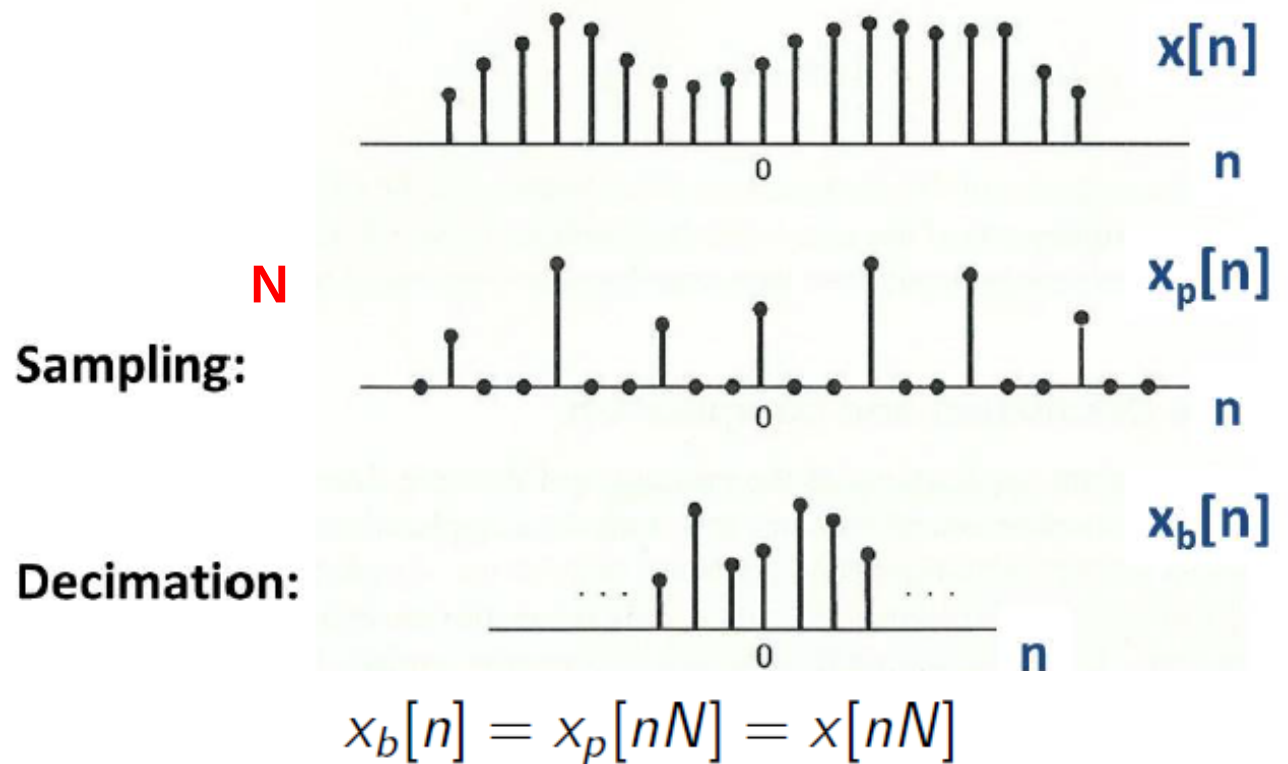


- How can we design a CT LTI system with frequency response H_C via DT LTI system?
- Step 1: Sampling frequency ω_s or $2\pi/T$ should be larger than Nyquist rate
- Step 2: $\tilde{H}_D(e^{j\omega T}) = H_C(j\omega)$
- Step 3: Frequency response of DT LTI system
 $H_D(e^{j\omega T}) = \sum_{k=-\infty}^{\infty} \tilde{H}_D(e^{j(\omega - k\omega_s)T})$ or
 $H_D(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \tilde{H}_D(e^{j(\omega - k\omega_s T)}) = \sum_{k=-\infty}^{\infty} H_C(j\frac{\omega - 2k\pi}{T})$

Review

Decimation

- After sampling, there will be a great amount of redundancy
- **Decimation**: discrete-time sampling + remove zeros



What is the relation between $X(e^{j\omega})$ and $X_b(e^{j\omega})$?

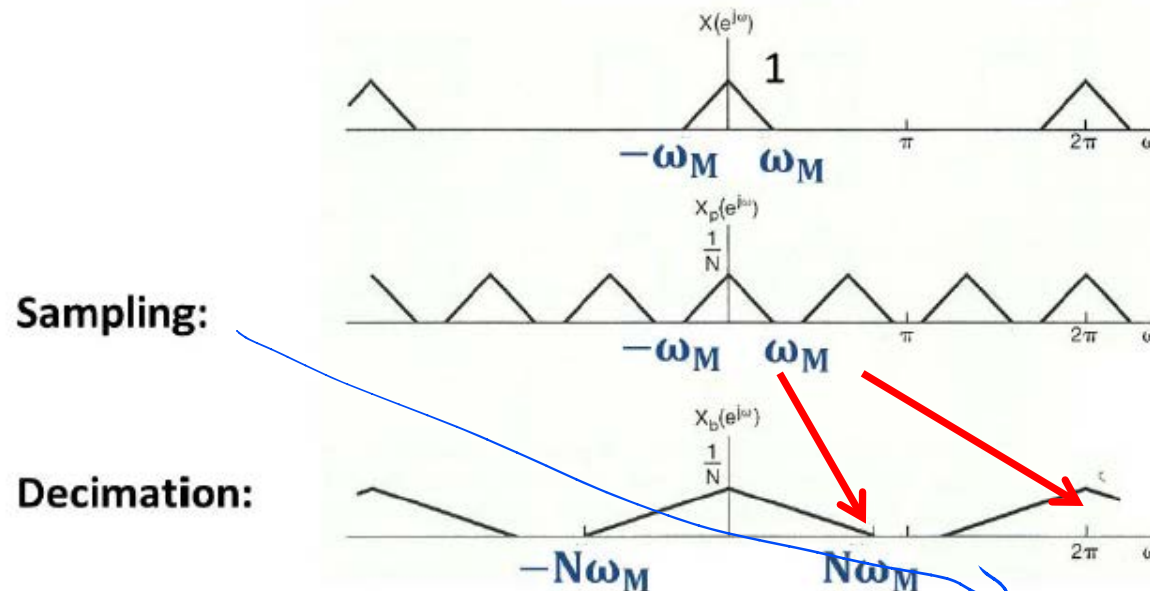
Review

Frequency Analysis

$$\begin{aligned}
 X_b(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_b[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N}) \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j\omega/N - k\omega_s})
 \end{aligned}$$

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$

$$\omega_s = 2\pi/N$$



$$\omega_s = 2\pi$$

$$\omega_s = \frac{2\pi}{N}$$

$$\omega_s = 2\pi$$

$$2\omega_M < \omega_s$$

Condition for Perfect Reconstruction: $N\omega_M < \pi$

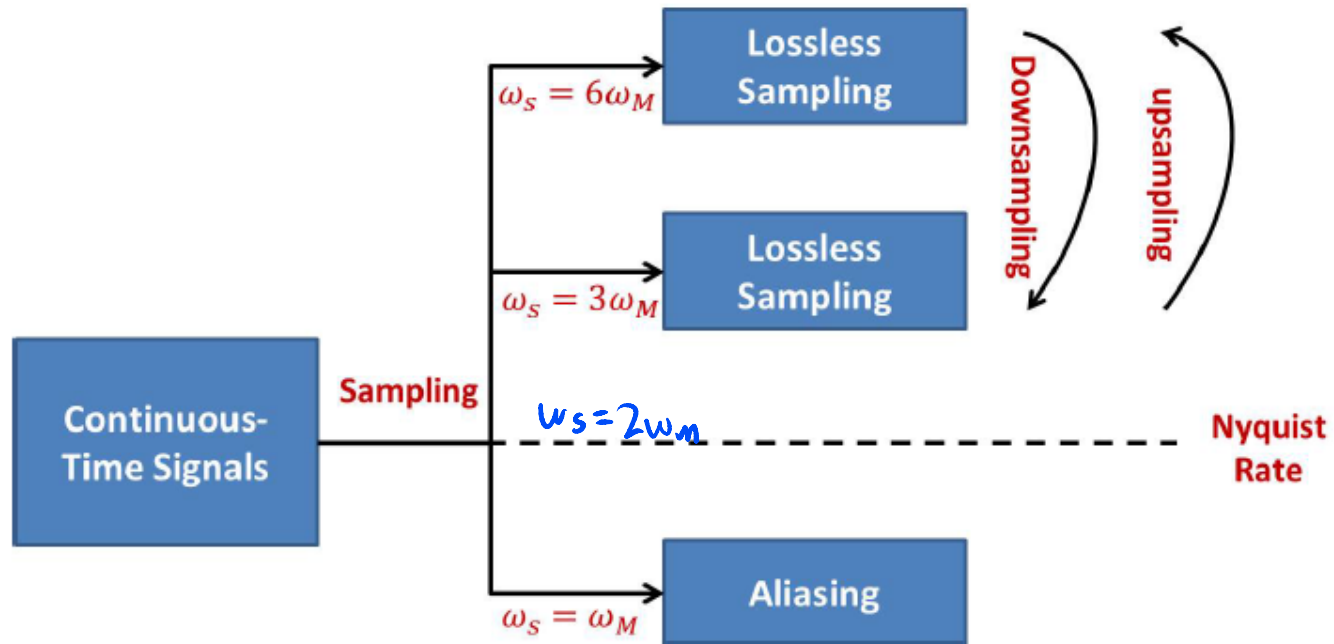
不重叠

Review

Anything Else?

改变采样频率

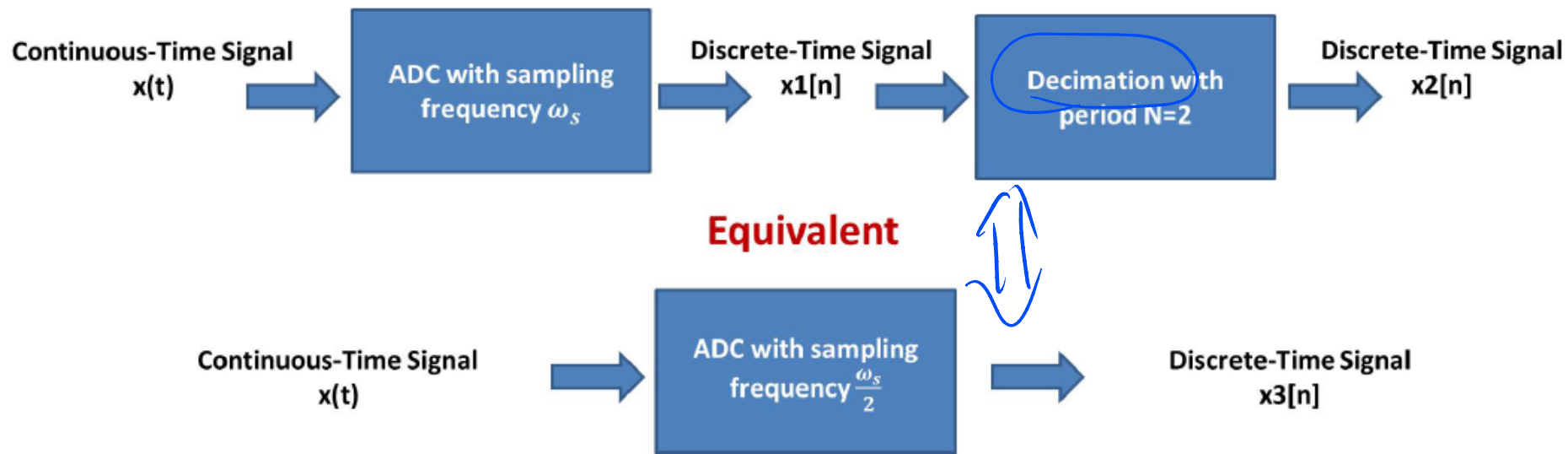
9



- **Downsampling**: to reduce the sampling frequency (decimation)
- **Upsampling**: to generate a DT signal with higher sampling frequency
- As long as Nyquist rate is satisfied, the transform between low-sampling-frequency version and high-sampling-frequency versions is lossless.

Downsampling

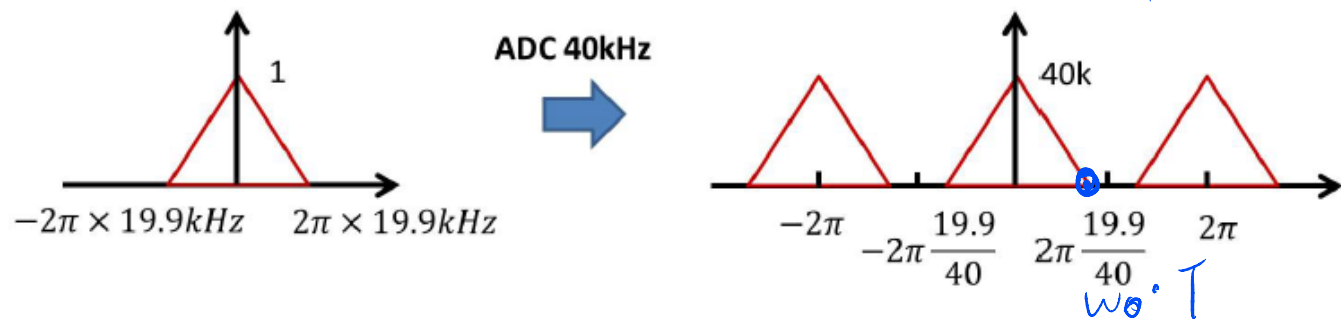
- Downsampling: a general procedure to reduce the sampling frequency



When do we need downsampling?

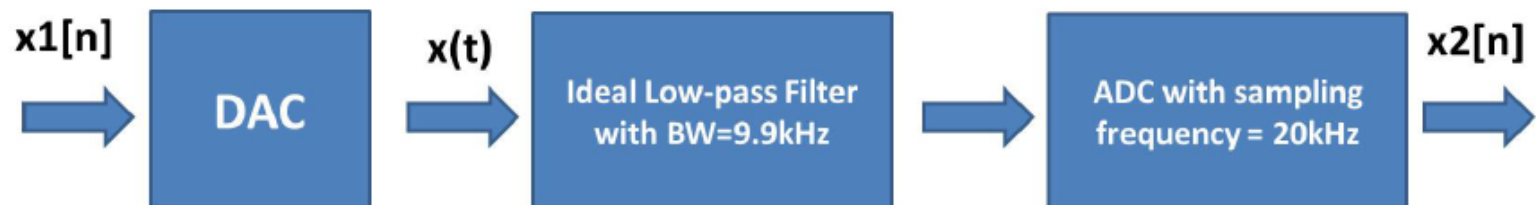
Downsampling Example (1/2)

- Suppose we have a clip of voice, $x(t)$, with bandwidth = 19.9kHz
- It can be converted to DT signal with sampling frequency 40kHz, denoted as $x_1[n]$



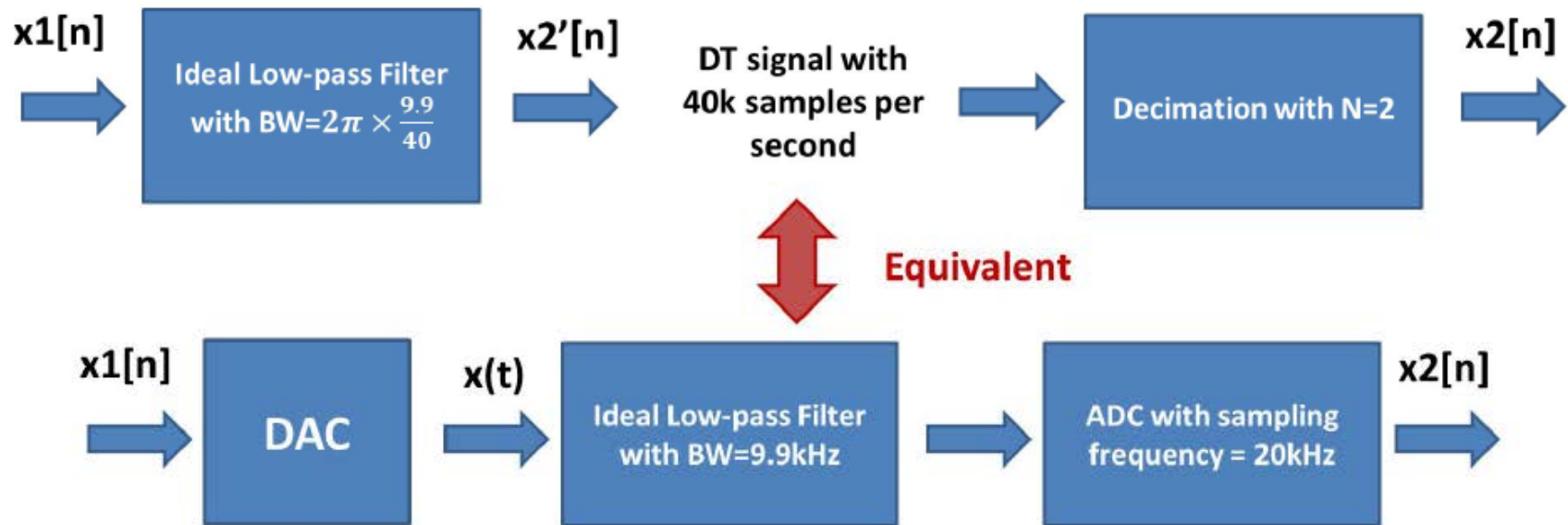
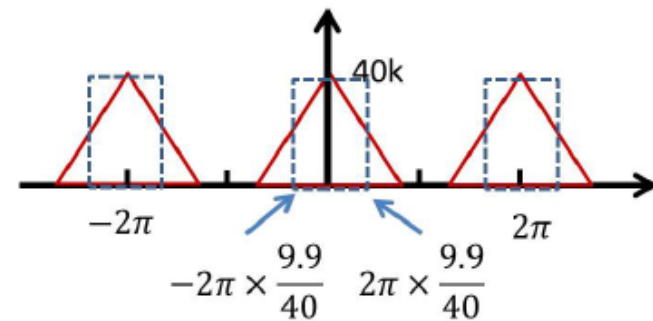
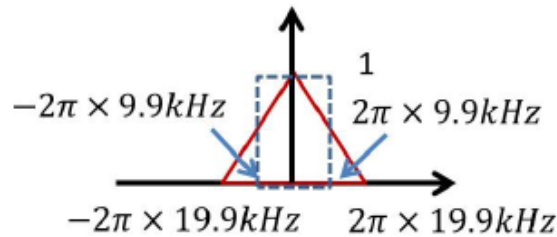
- Based on $x_1[n]$, if we want to save the voice information within 9.9kHz into another DT signal, what can we do?

One Choice:



How can we generate $x_2[n]$ in discrete-time domain?

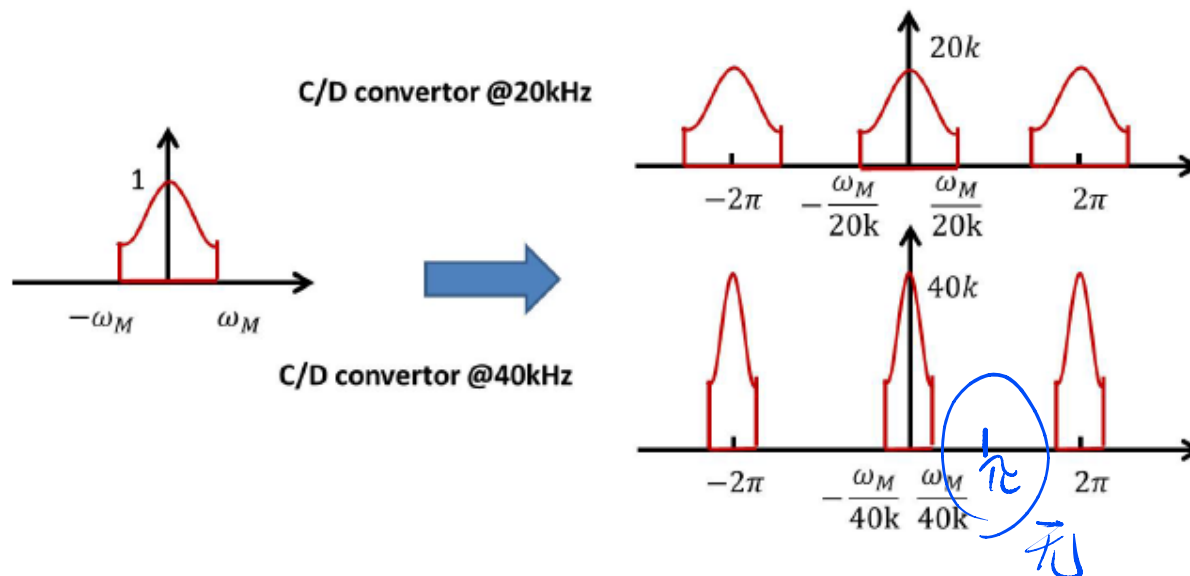
Downsampling Example (2/2)



Review

Upsampling

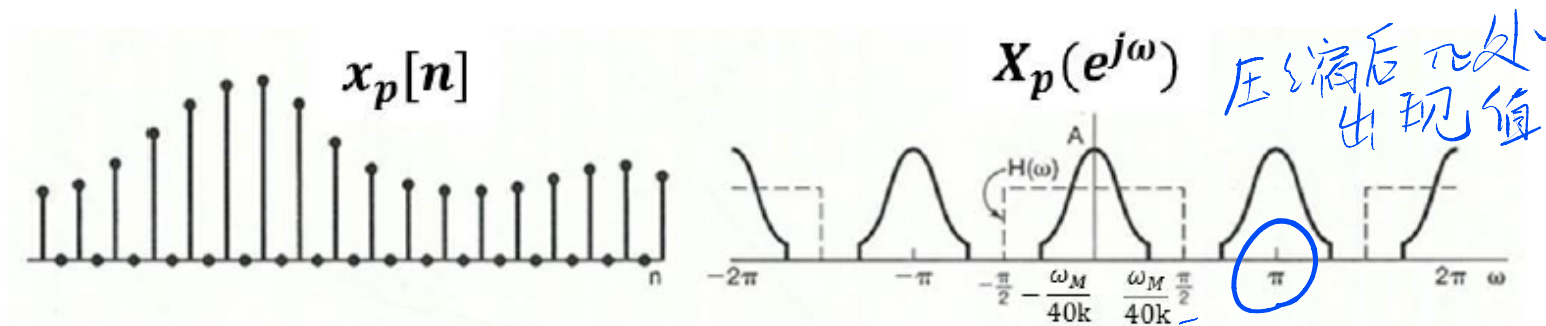
- **Upsampling:** a procedure to generate a sequence with higher sampling frequency
- Superpose the following two digital sound clips
 - ▶ Audio clip 1: Bandwidth= 19.9kHz, sampled at 40kHz
 - ▶ Audio clip 2: Bandwidth= 9.9kHz, sampled at 20kHz
- Double the sampling frequency of audio clip 2 (40kHz)
- How to do upsampling in discrete-time domain?



Review

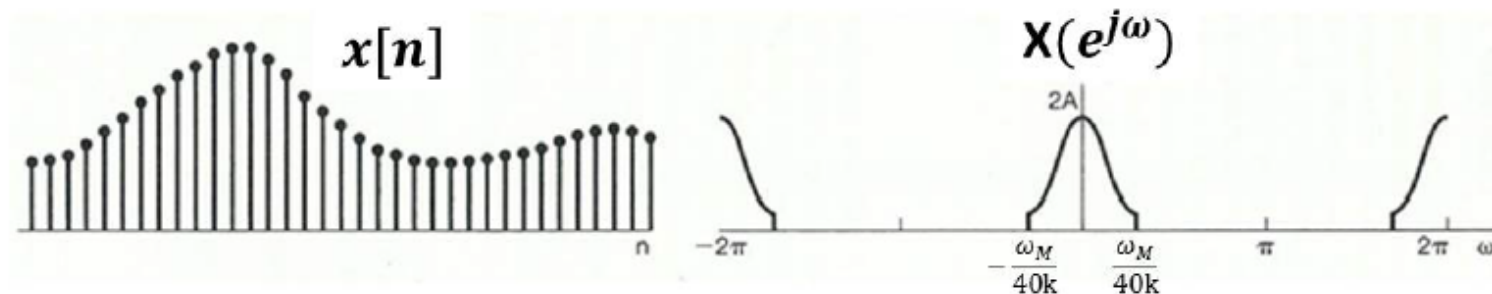
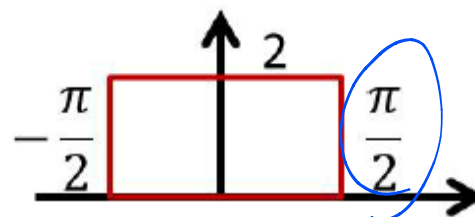
- Time expansion (Insert zeros):

$$x_p[n] = x_{b(k)}[n] \longleftrightarrow X_p(e^{j\omega}) = X_b(e^{jk\omega})$$



- Low-pass filtering:

Ideal Low-Pass Filter:



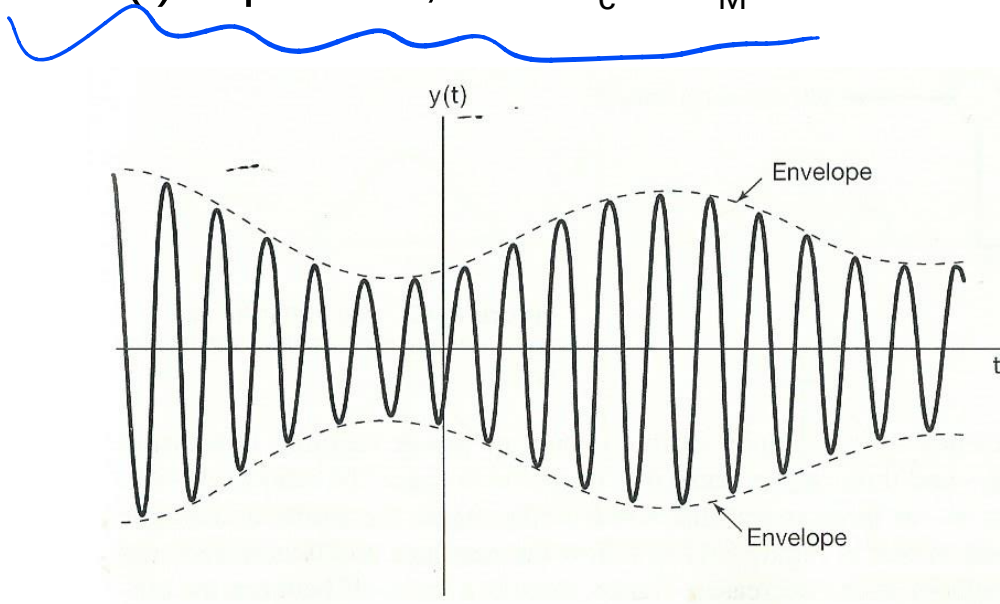
Summary

- Meaning of amplitude modulation
 - ◆ with a complex exponential carrier
 - ◆ with a sinusoidal carrier
- Demodulation for sinusoidal AM
 - ◆ Synchronous demodulation 同步解调 (乘载波信号, 频率相位相同)
 - ◆ Asynchronous demodulation, and its two important assumptions 异步 (包络: 整流 + 低通) 同频
- Frequency-division multiplexing (FDM)

Asynchronous demodulation

- Avoid the need for synchronization between modulator and demodulator.

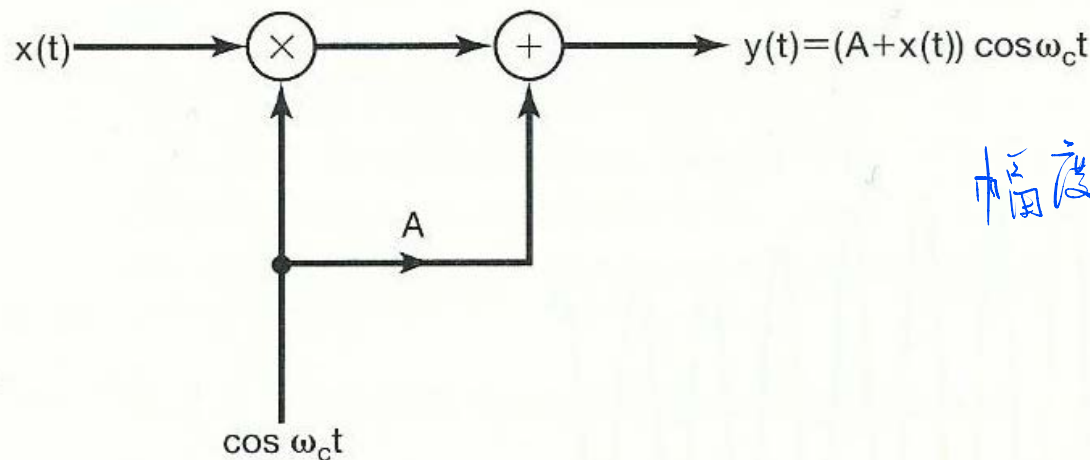
An example of modulated signal $y(t)$, when $x(t)$ is positive, and $\omega_c > \omega_M$.



Envelope signal, a smooth curve connecting the peaks, approaches to the modulating signal $x(t)$.

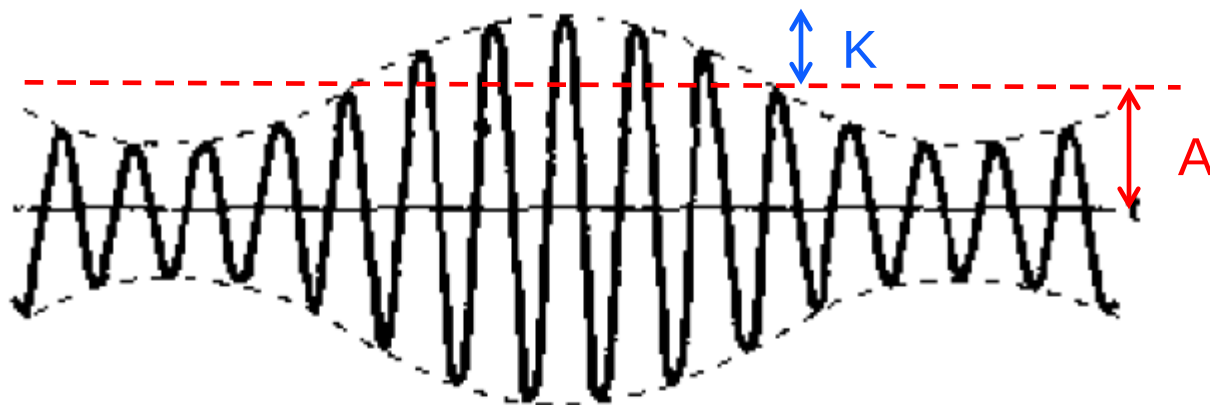
AM in an asynchronous system

- Two important assumptions:
 - ω_c is higher than ω_M
 - $x(t)$ is always positive



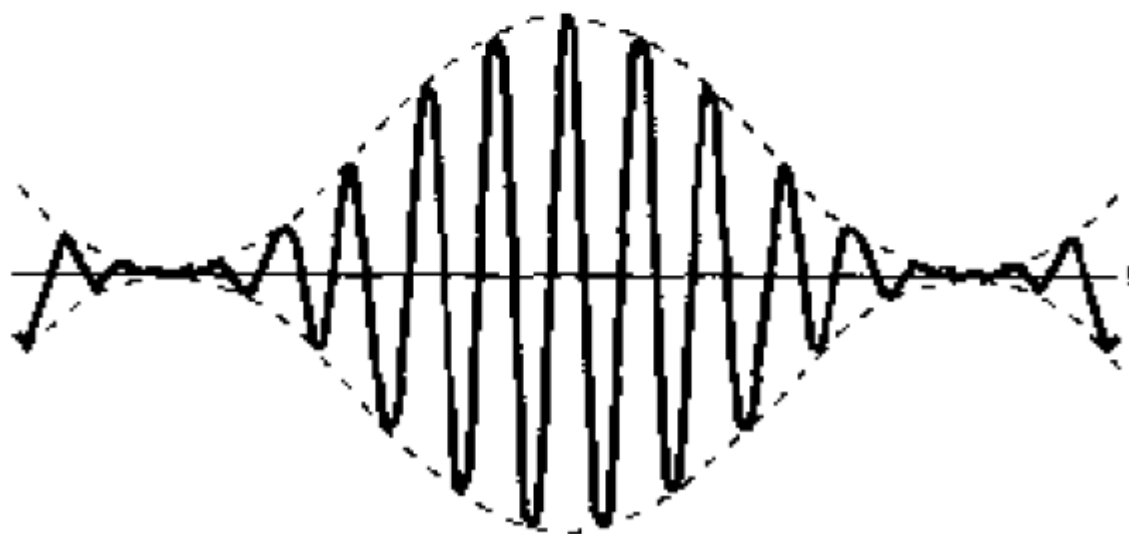
1) Add a constant A to make sure $[A+x(t)]$ is positive. Let K be the maximum amplitude of $x(t)$, or $|x(t)| \leq K$. We need to have $A > K$.

2) Define K/A as **modulation index m** , which is from 0 to 1.



$m=0.5$

(a)



$m=1$

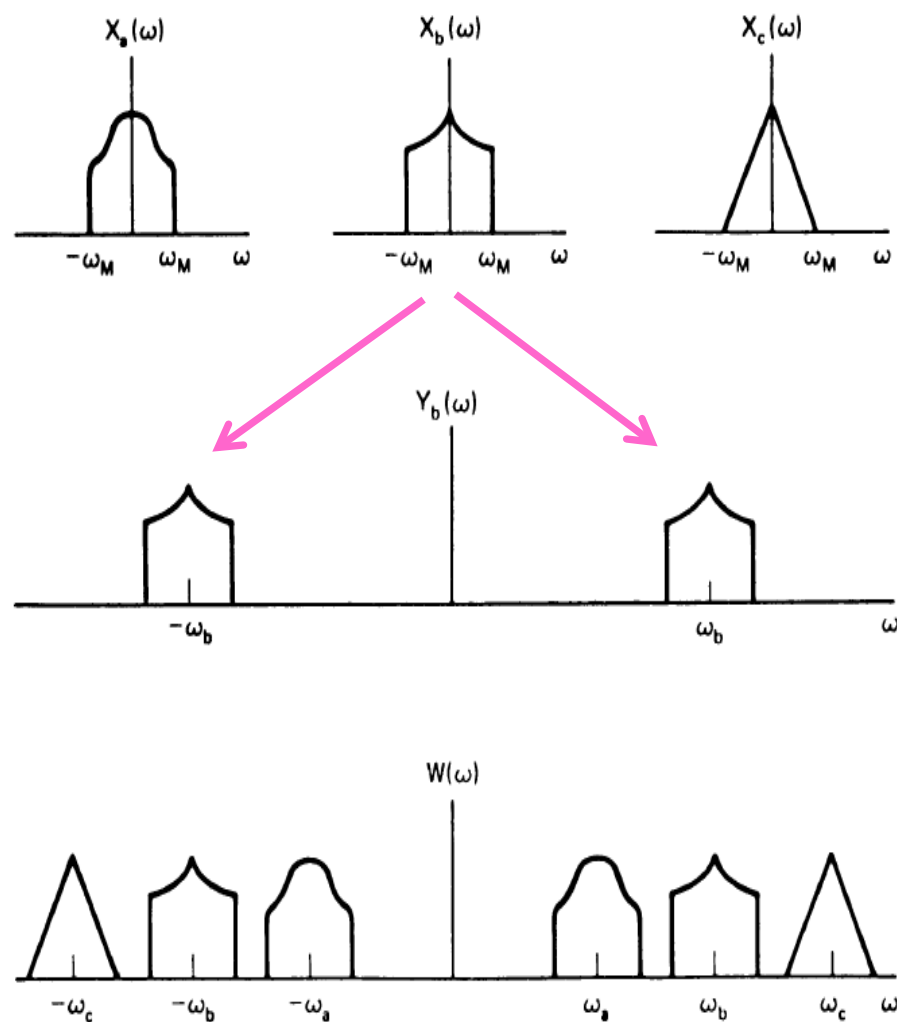
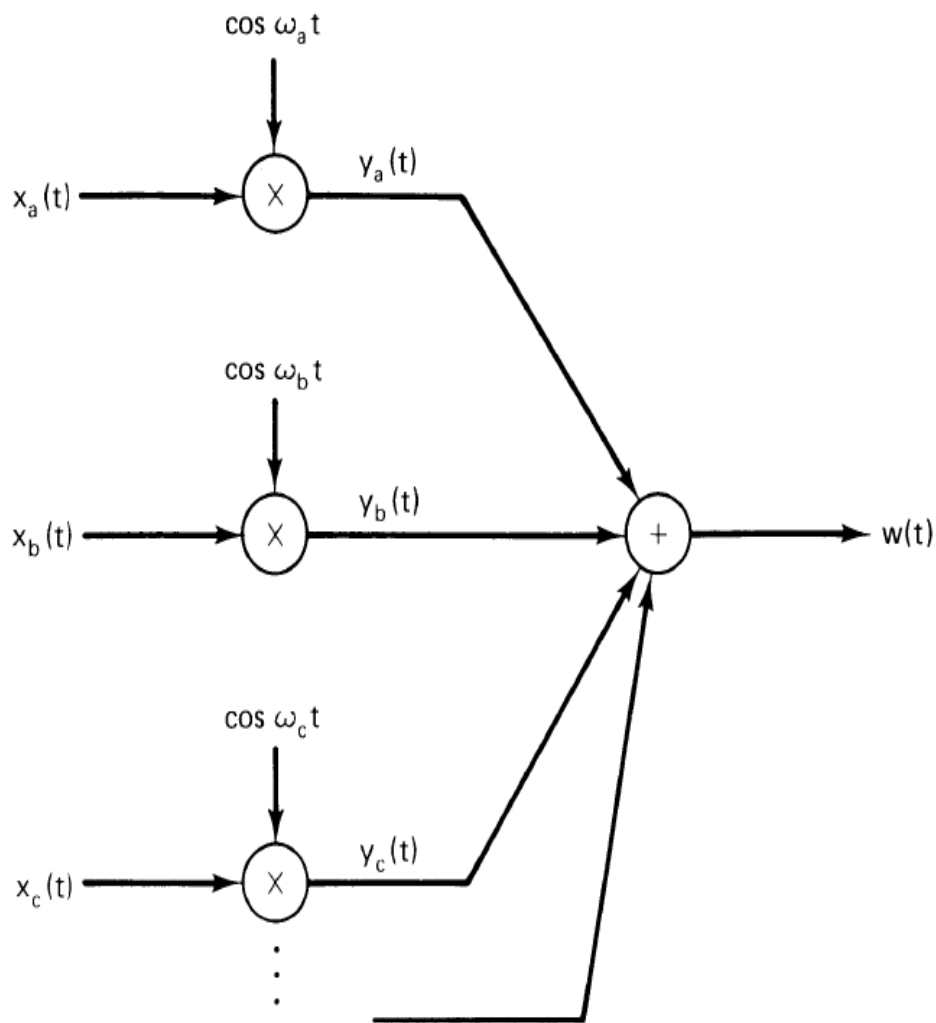
(b)

What about $m=0$?

Frequency-division multiplexing (FDM)

- Systems for transmitting signals provide more bandwidth than is required for one signal.
 - ◆ E.g., speech signal → 20 ~ 20 kHz
 - microwave channel → 300 MHz ~ 300 GHz
 - satellite link → a few hundred MHz ~ 40 GHz
 - (more in Fig. 8.18)
- Different modulating signals (e.g., speech), which are overlapping in frequency, can have their spectra shifted (e.g., by sinusoidal AM) without overlapping, so they can be transmitted simultaneously over a single wideband channel.

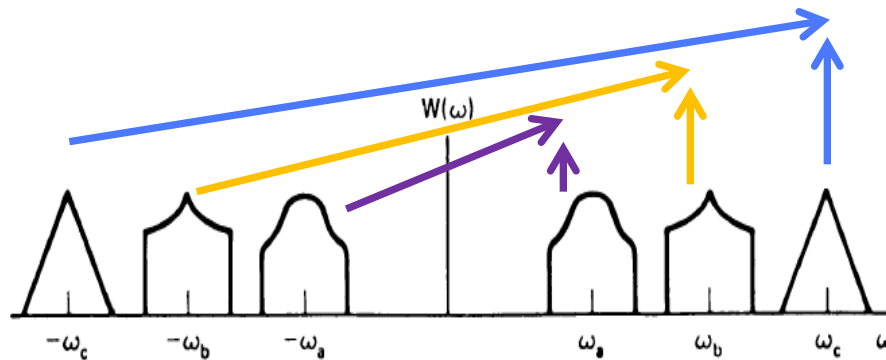
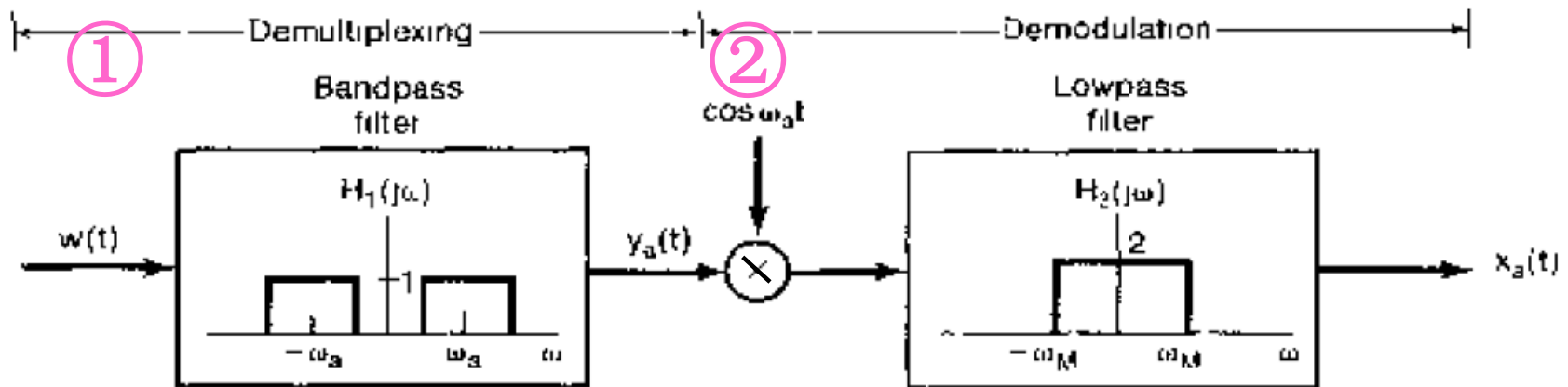
FDM (cont.)



Review

FDM demultiplexing and demodulation

- ① bandpass filtering to have the modulated signal from one channel
- ② demodulation to recover the modulating signal



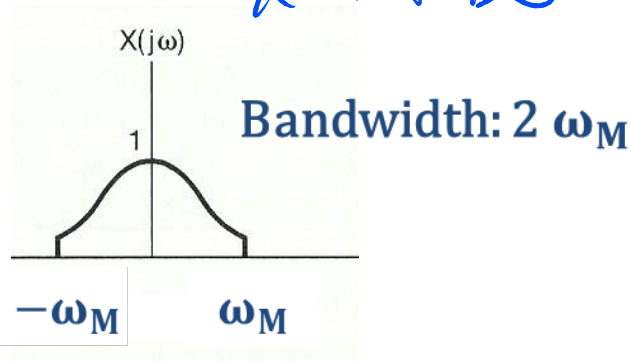
Occupy twice the original bandwidth
 → insufficient use of bandwidth

Review

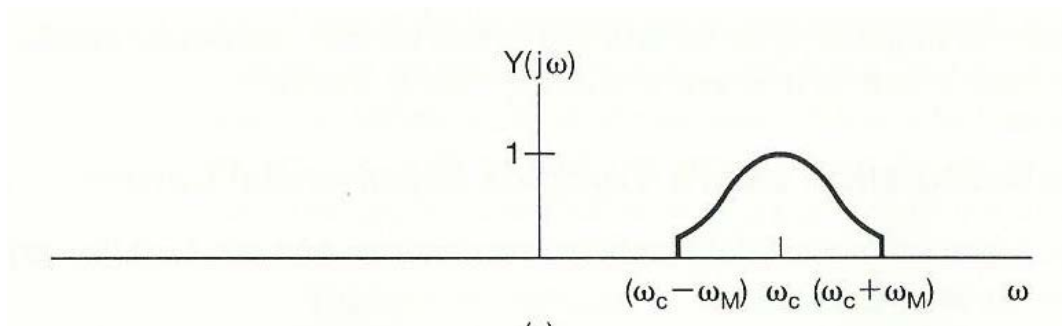
Single-sideband (SSB) sinusoidal AM

Occupied bandwidth:

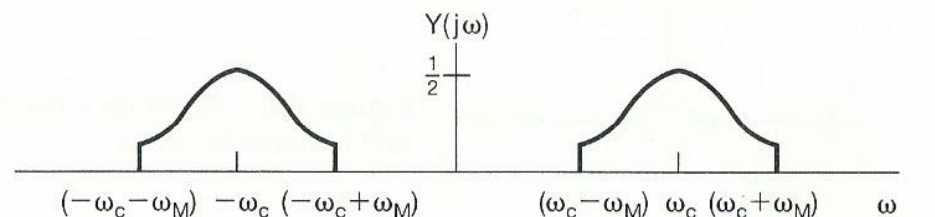
实偶信号



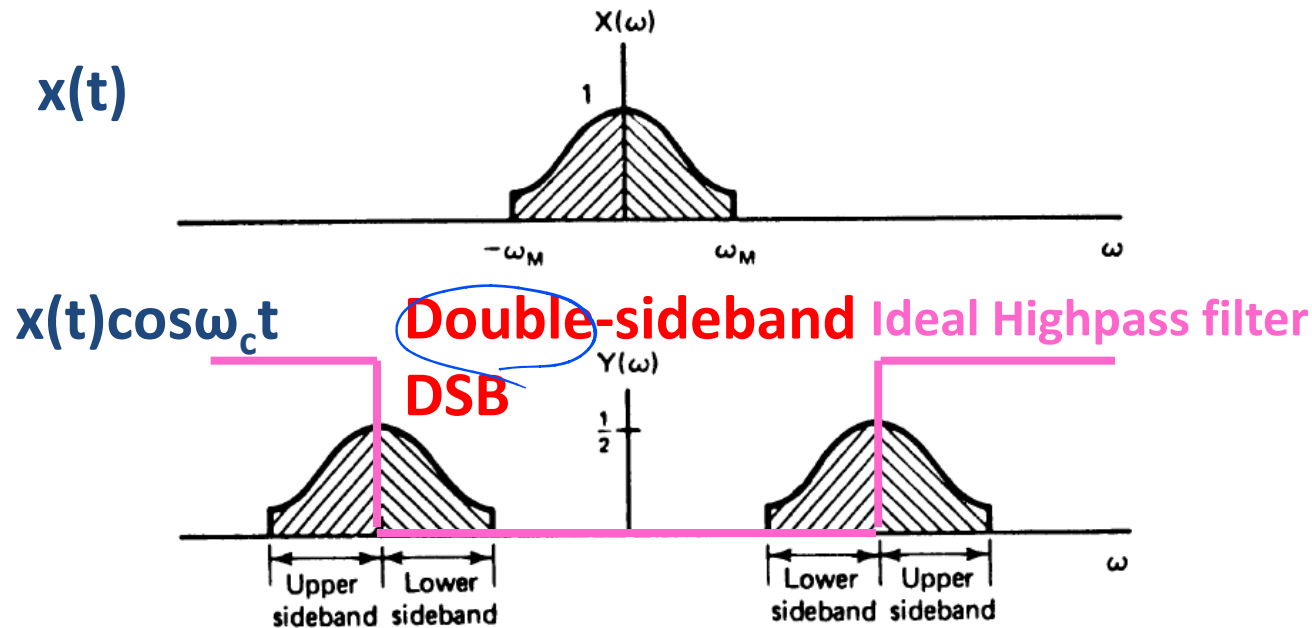
- With exponential carrier, the bandwidth is still $2\omega_M$



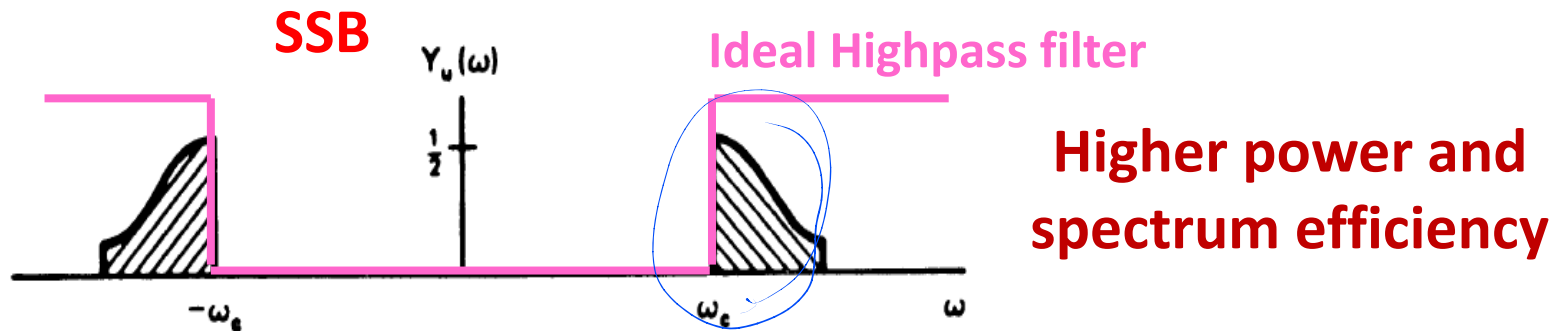
- With sinusoidal carrier, twice bandwidth is required.



SSB sinusoidal AM (cont.)



Observation: $x(t)$ can be recovered if two upper (or lower) sidebands are retained.



Sinusoidal AM: Summary

- Four types of sinusoidal AM
 - ▶ AM-DSB/SC: $y(t) = x(t)\cos\omega_c t$
 - ▶ AM-DSB/WC: $y(t) = (x(t) + A)\cos\omega_c t$
 - ▶ AM-SSB/SC: AM-DSB/SC + ideal highpass/lowpass filter
 - ▶ AM-SSB/WC: AM-SSB/SC + $A\cos\omega_c t$

WC: with carrier

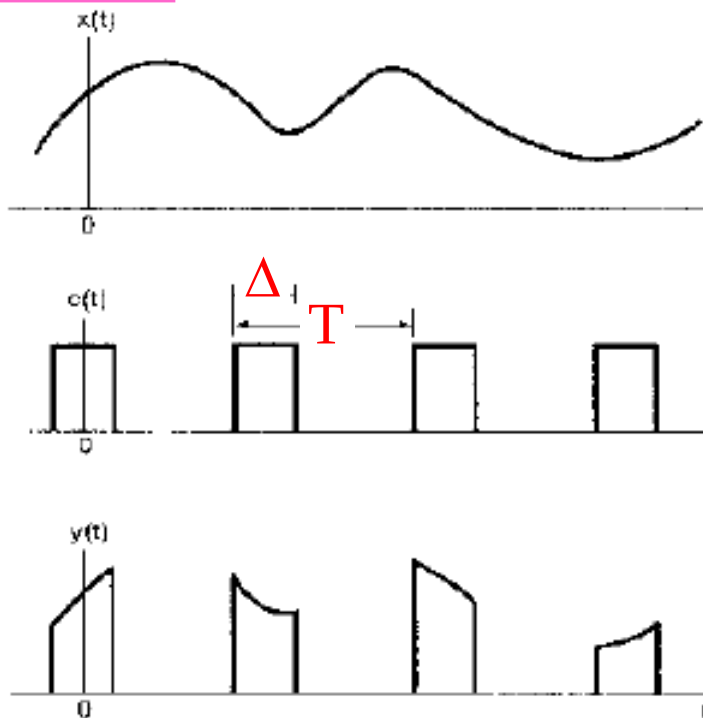
$$y(t) = \underbrace{\sin}_{\text{sin}} \omega_c t + m(\sin \omega_m t)(\sin \omega_c t).$$

WC

Amplitude modulation with a pulse train

- Carrier signal could be a sinusoidal signal, or a pulse train.

Figure 8.23



$$X(j\omega)$$

$$C(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_c)$$

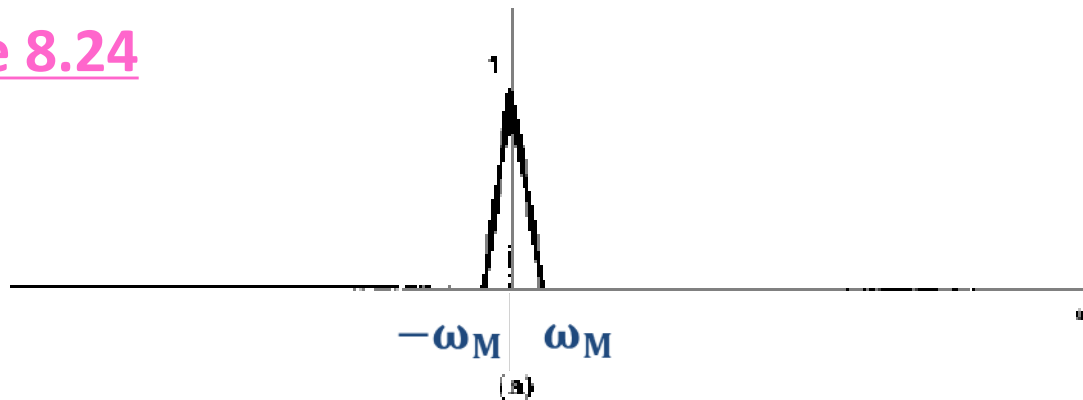
$$a_k = \frac{\sin(k\omega_c \Delta / 2)}{\pi k}$$

[Read Example 3.5](#)

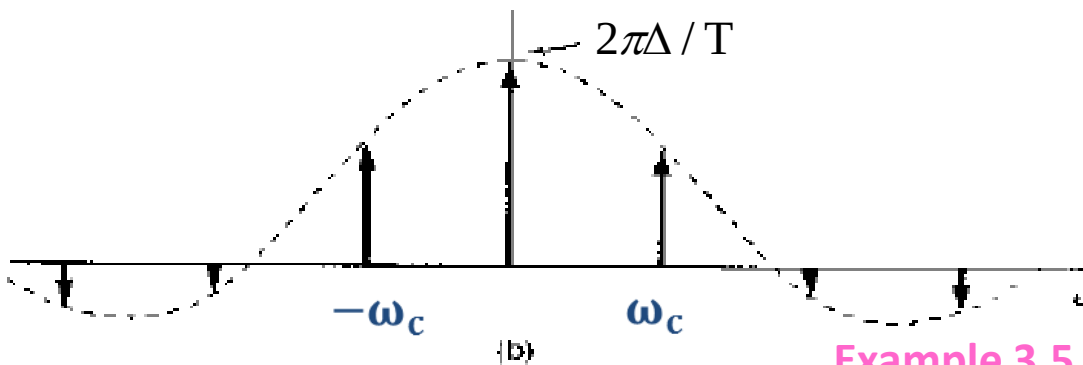
$$Y(j\omega) = \sum_{k=-\infty}^{\infty} a_k X(j(\omega - k\omega_c))$$

$$X(j\omega)$$

Only affected by T



$$C(j\omega)$$

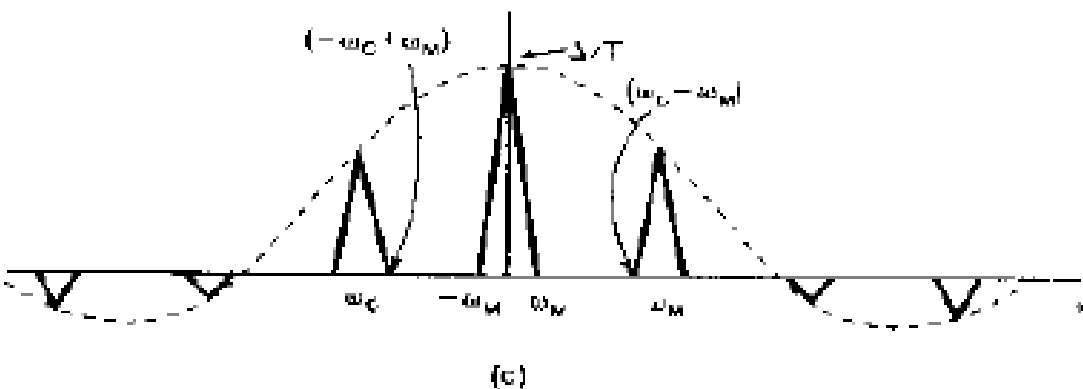


When $\omega_c > 2\omega_M$, $X(j\omega)$ can be recovered by lowpass filtering.

Note: similar to the condition in sampling.

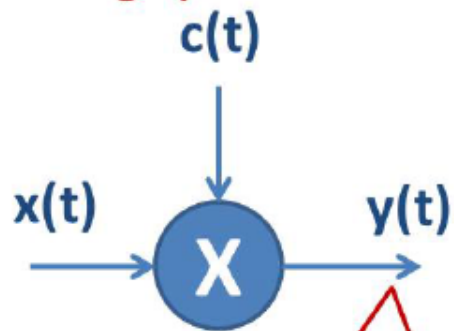
Example 3.5

$$Y(j\omega)$$

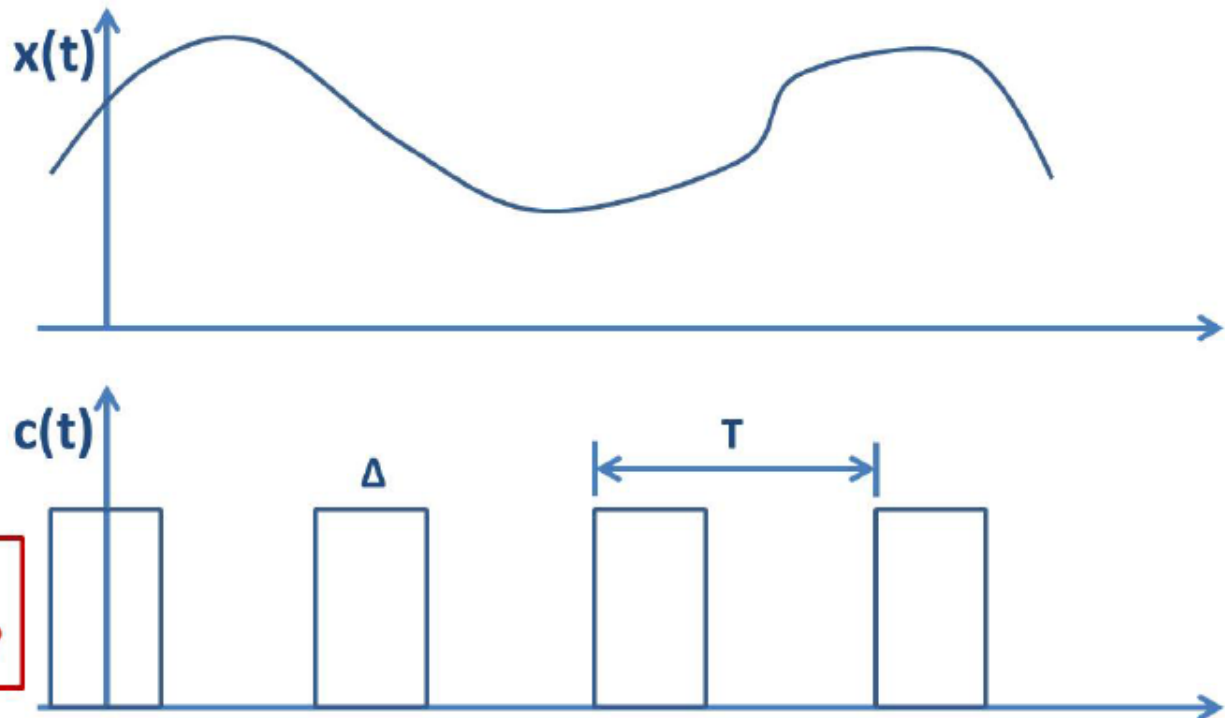


AM with Pulse-Train Carrier

Instead of COS, we use
rectangle pulse as carrier

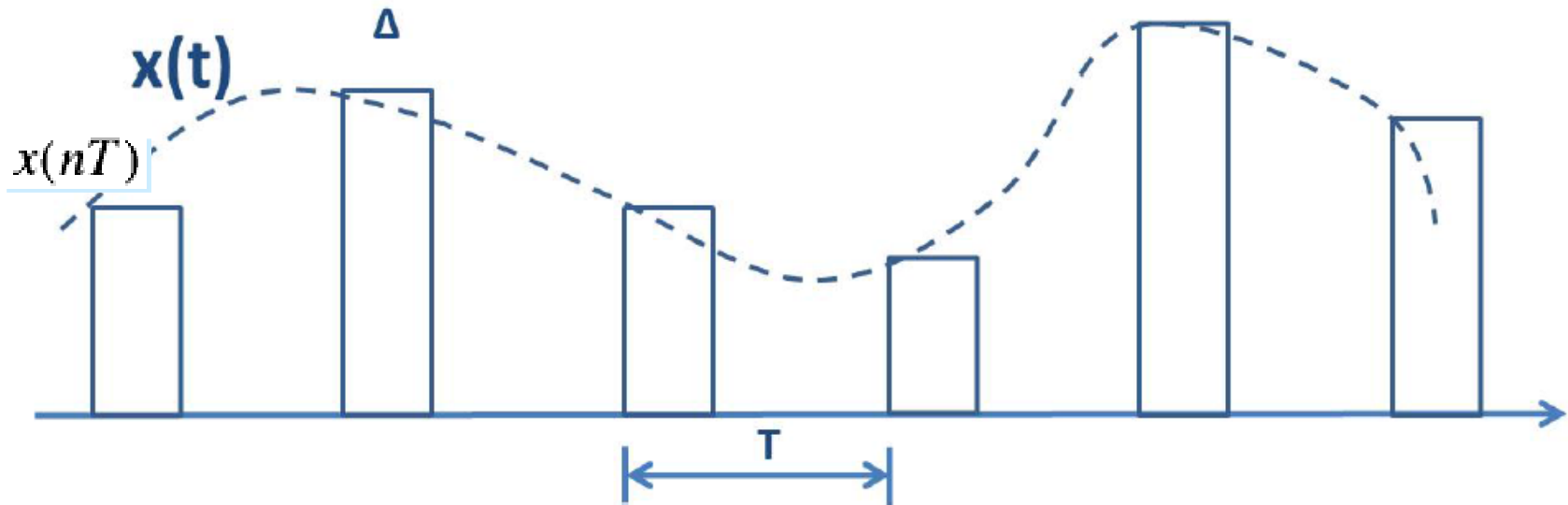


What's the shape of $y(t)$?
Can we demodulate x from y ?



$$C(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2 \operatorname{sinc} k \omega_0 \frac{\Delta}{2}}{k} \delta(\omega - k \omega_0) \quad \text{where} \quad \omega_0 = 2\pi / T$$

Pulse-Amplitude Modulation (PAM)



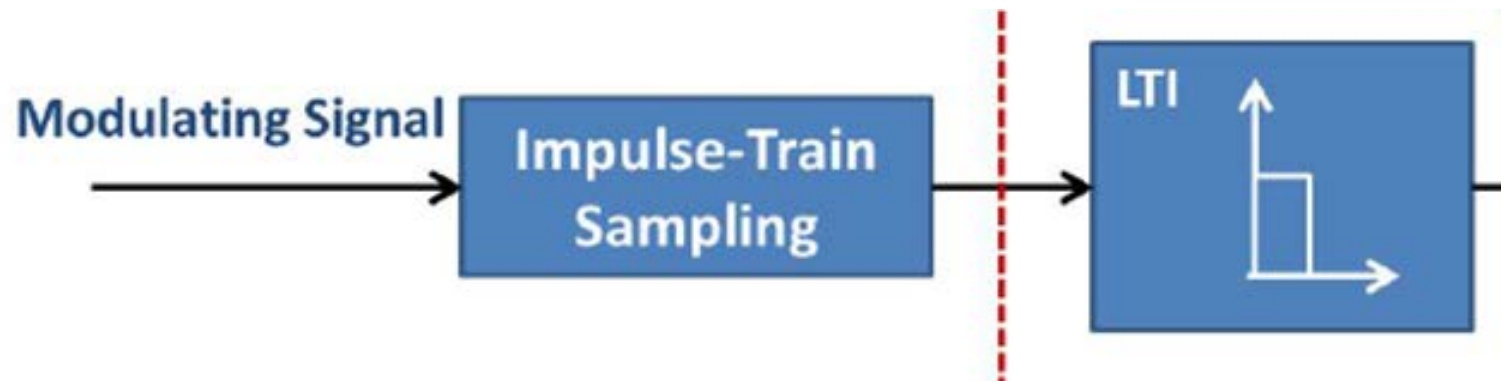
- Pulse-Amplitude Modulation (PAM): sample with period T and hold for duration Δ

保持 Δ

- Questions:

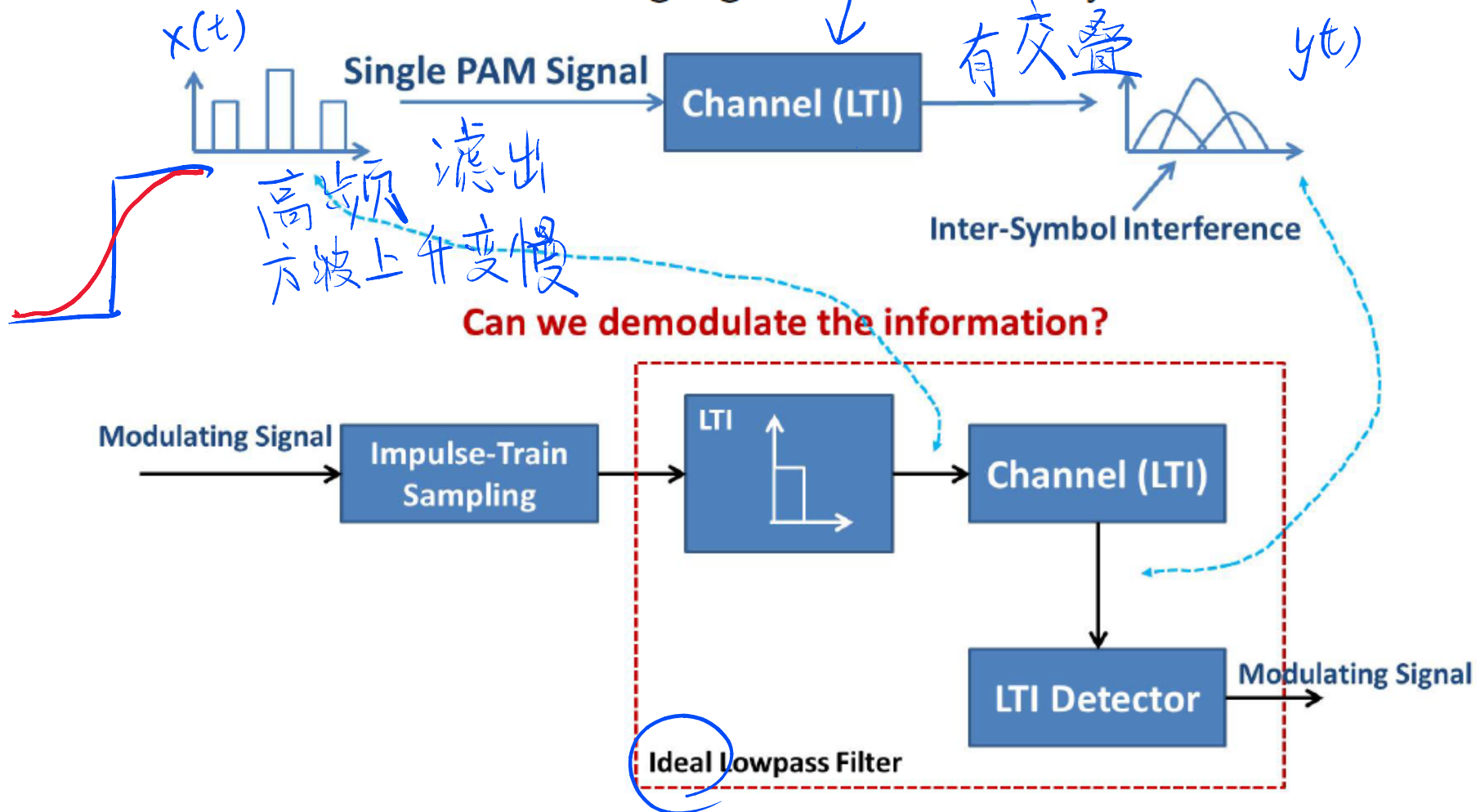
- ▶ Without channel distortion, how to demodulate?

Cont.



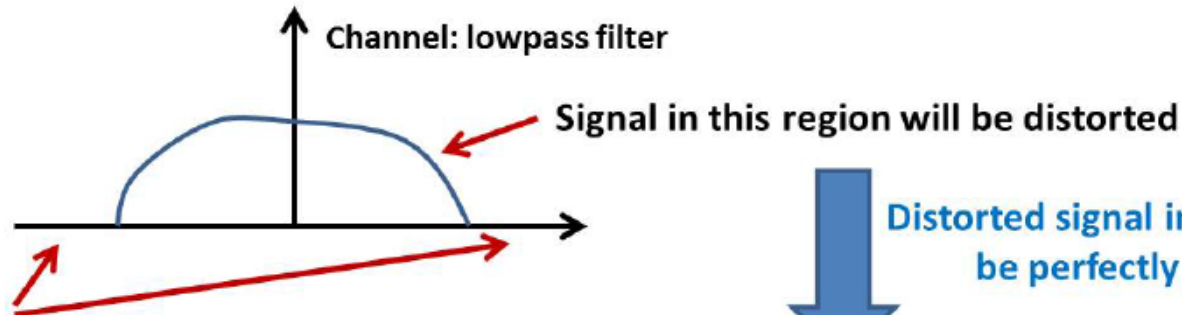
Inter-Symbol Interference (ISI)

- Communication channel is usually a low-pass filter
- Receiver recovers the modulating signal with an LTI system



Against ISI (1)

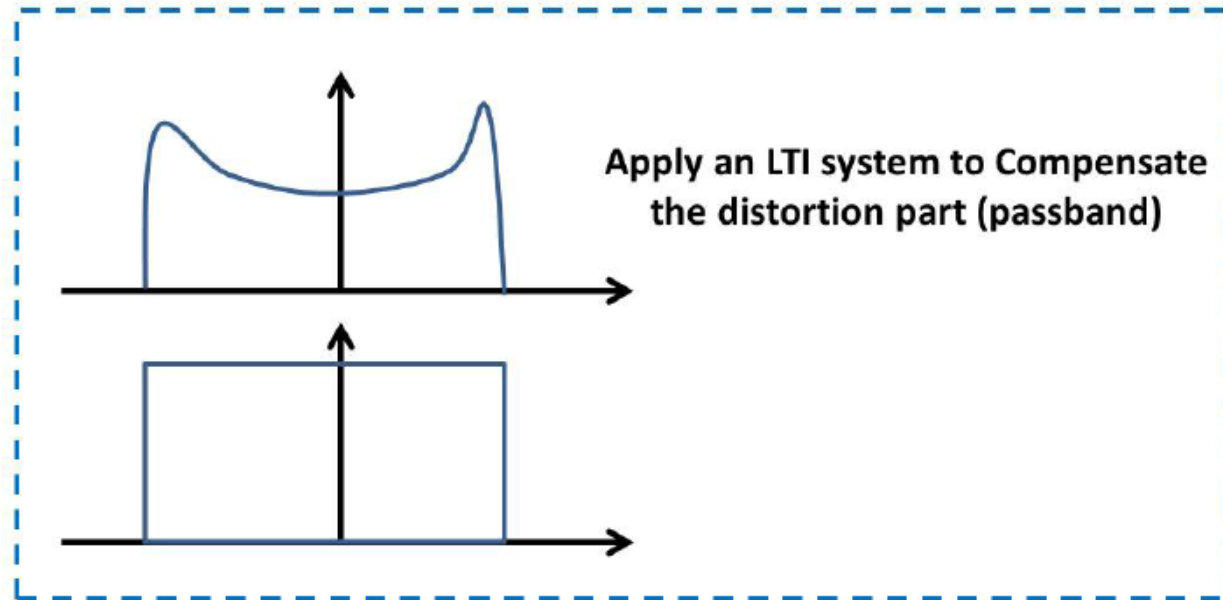
- What's the cause of ISI?



Distorted signal in this region can be perfectly recovered

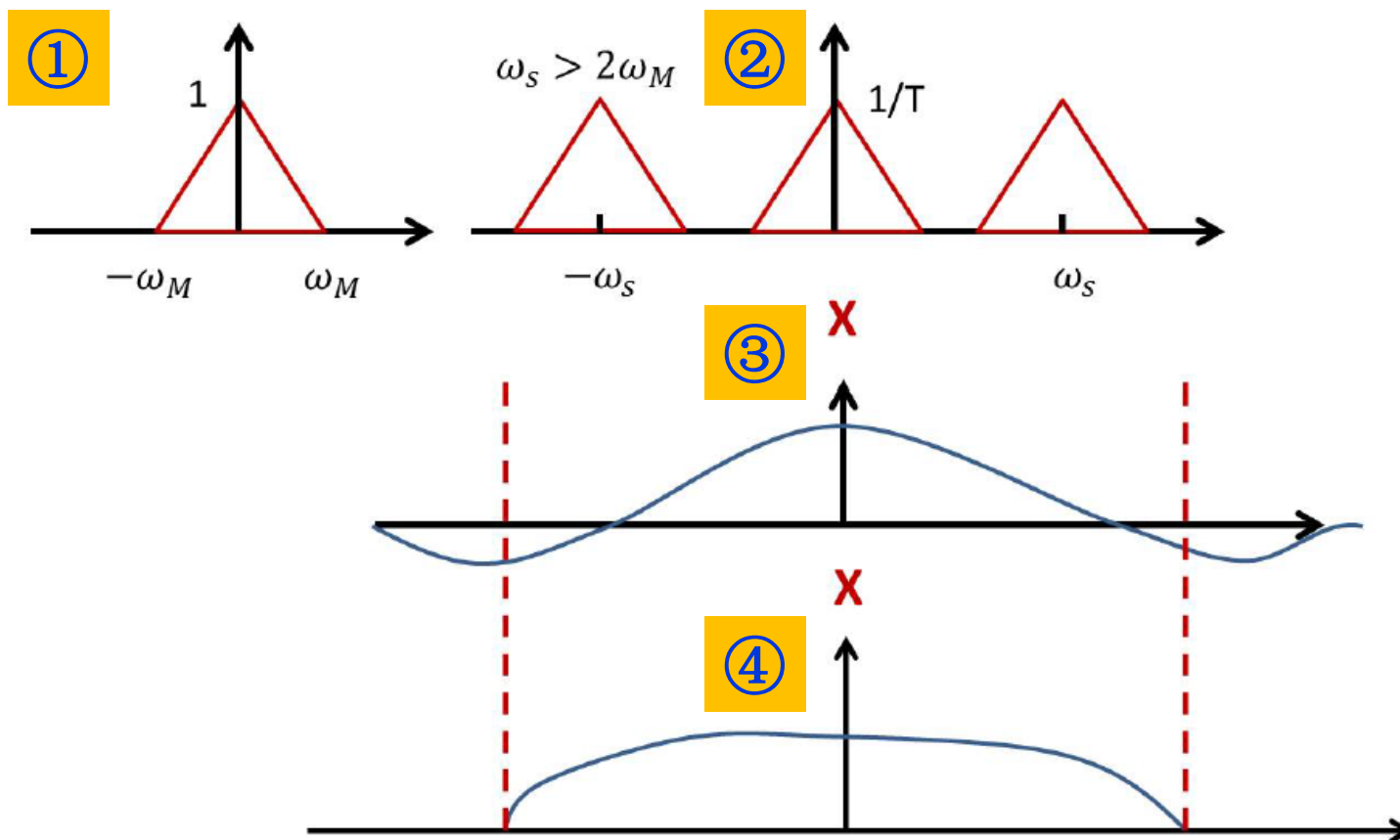
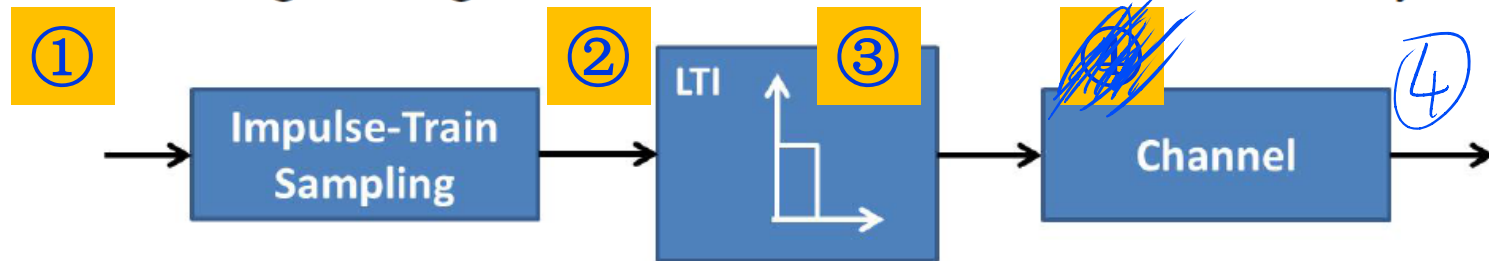
Signal in this region will be eliminated

After channel, signal information in this region will disappear



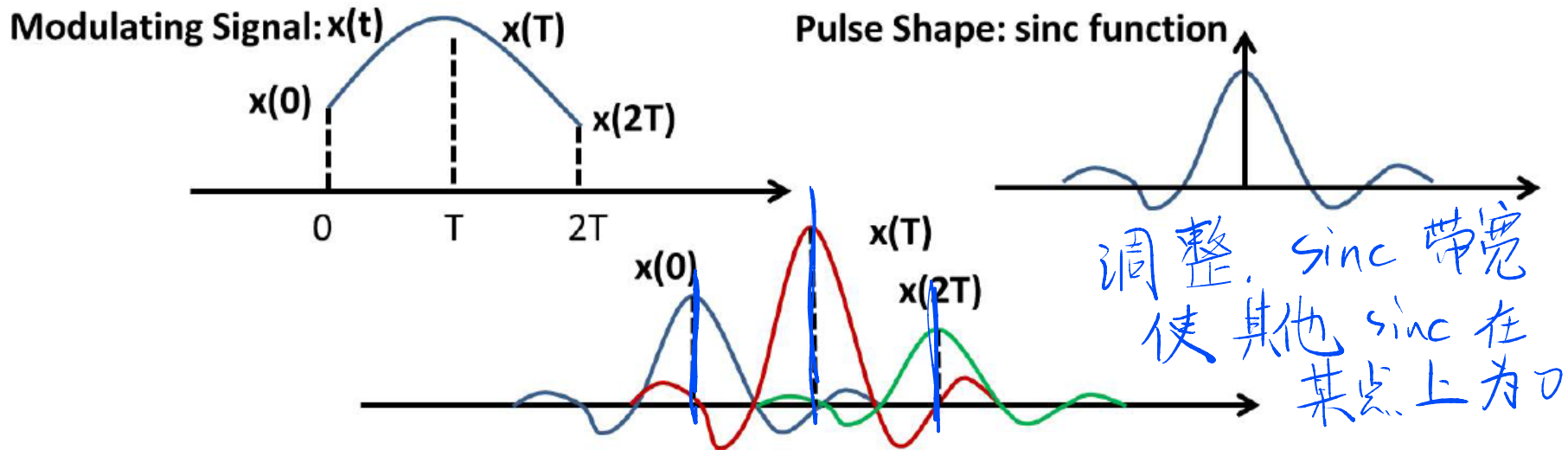
Against ISI (2)

- When using rectangular wave in PAM, the bandwidth is infinity

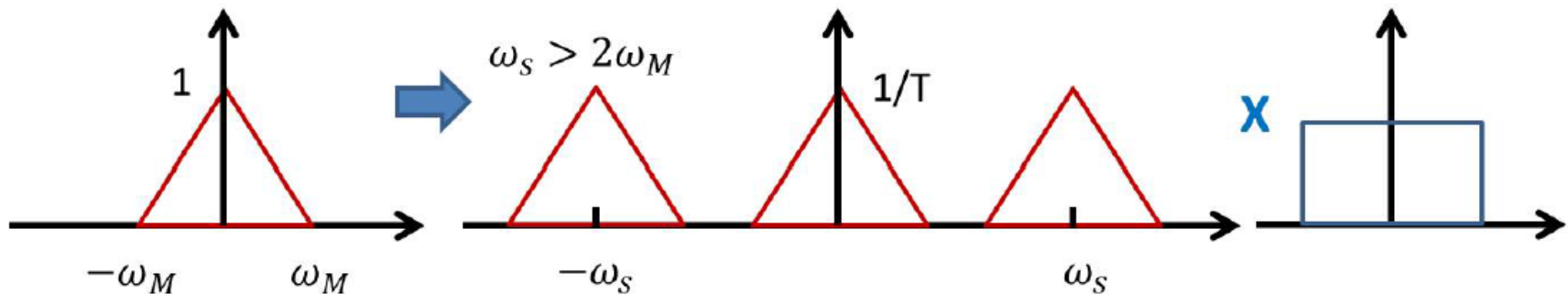


Against ISI (2) (Cont.)

- To avoid ISI, PAM signal should use band-limited pulse. E.g., sinc function



1. Receiver can perfectly recover this signal
2. Sampled values are kept in the peaks of pulses



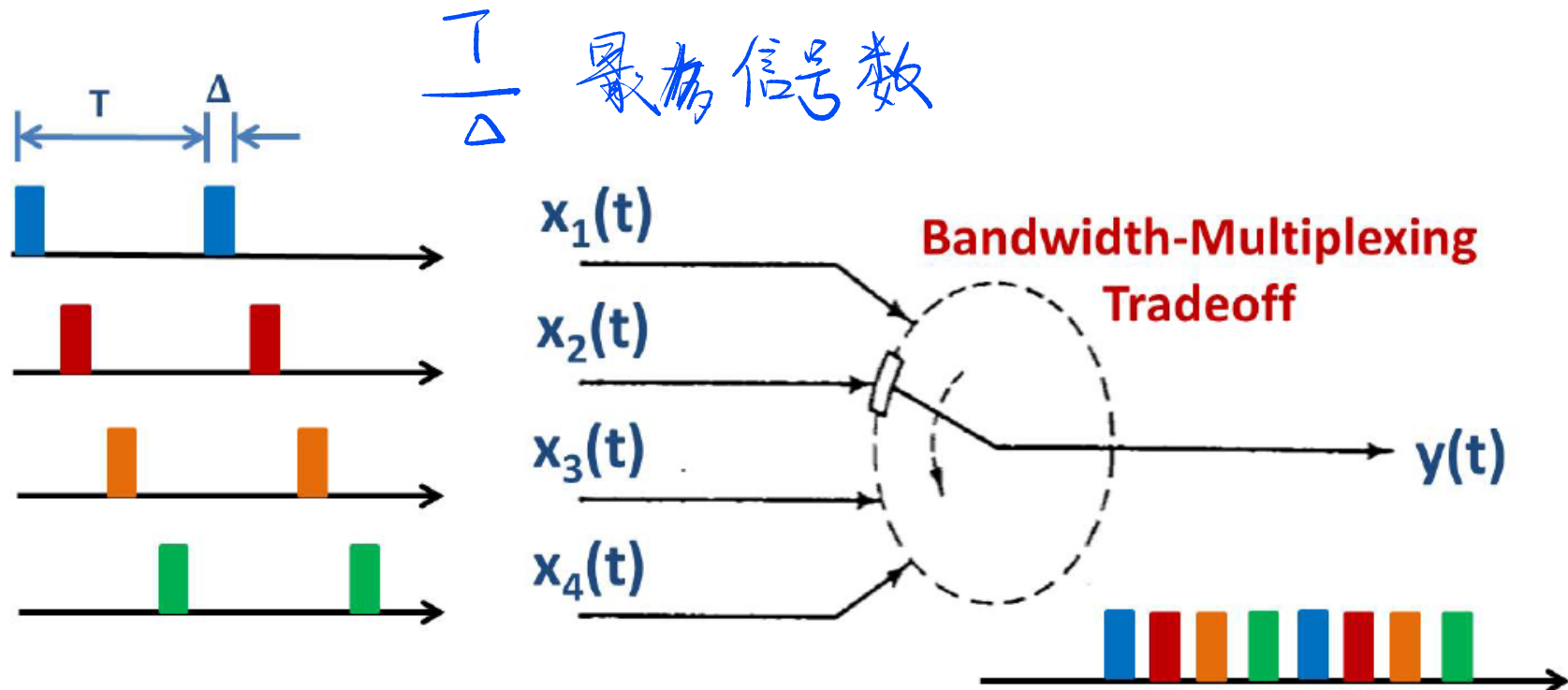
Summary on PAM

- Modulation $\leftrightarrow$ Sampling (zero-order hold); Demodulation $\leftrightarrow$ Recovery
- In addition to rectangular wave, other pulses can be used for PAM
- Demodulation without channel distortion
- Demodulation with channel distortion (ISI)
 - ▶ Recover modulating signal: make sure pulse shaping + channel + receiver's detector = ideal low-pass filter
 - ▶ Recover modulated signal: use band-limited wave in PAM. E.g., sinc wave

Time-Division Multiplexing (TDM)

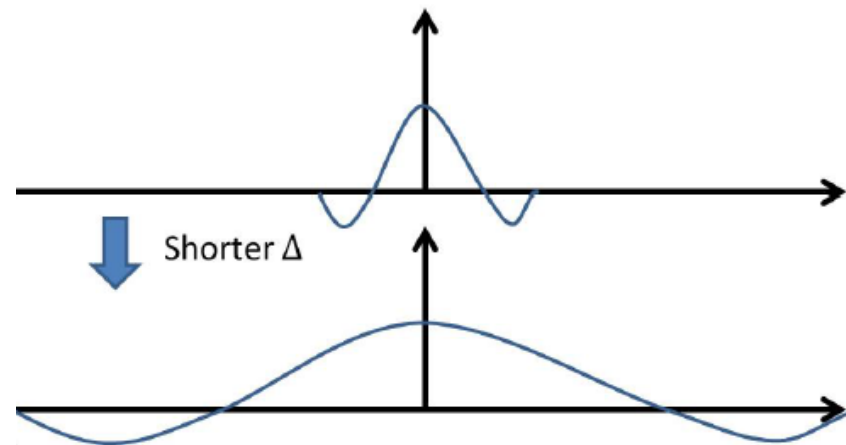
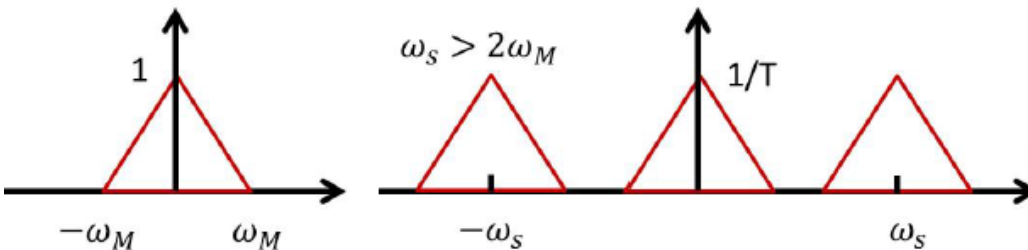
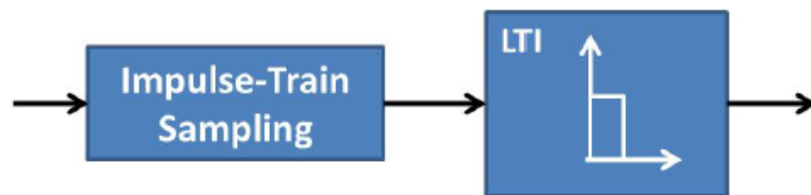
时域不重叠

AM signals with pulse-train carrier or PAM signals can be multiplexed in time domain



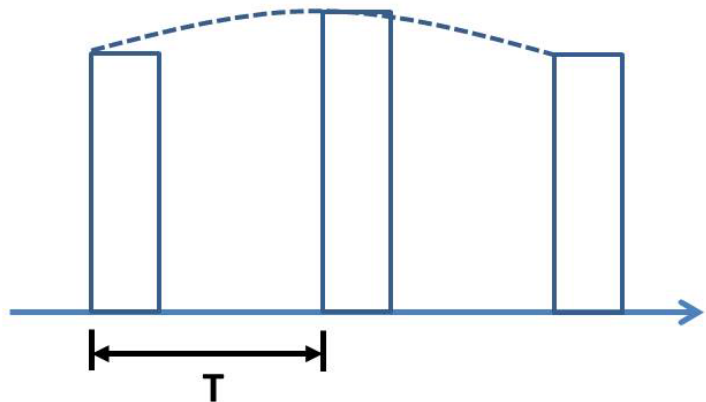
Cont.

- The number of multiplexed signals is determined by T and Δ
- Can T be as large as we want?
 - ▶ T is the sampling period
 - ▶ Let ω_M be the bandwidth of modulating signal
 - ▶ $\frac{2\pi}{T} > 2\omega_M \Rightarrow T < \frac{\pi}{\omega_M}$
- Can Δ be as short as we want?
 - ▶ Shorter pulse $\Rightarrow$ larger bandwidth consumption

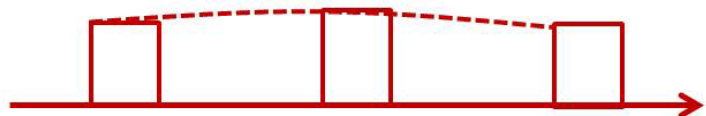


ISI in TDM

PAM Signal 1

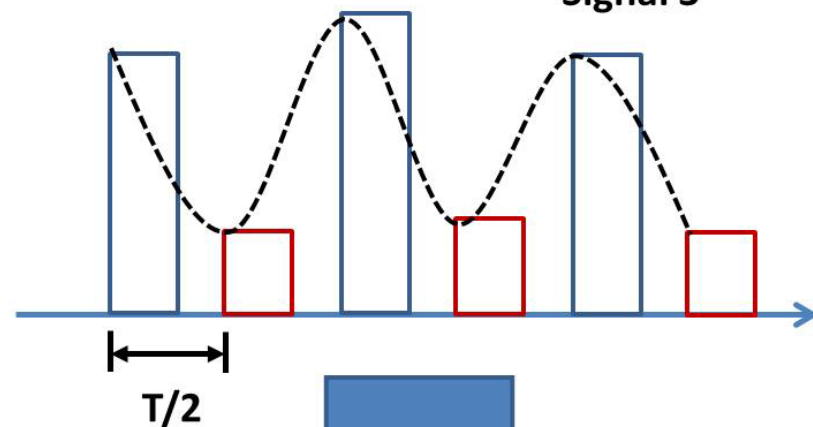


PAM Signal 2

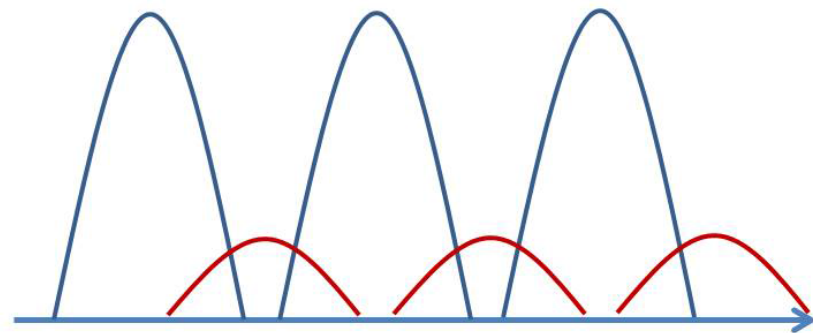


Similar to PAM, we have two approaches to handle ISI

TDM Signal



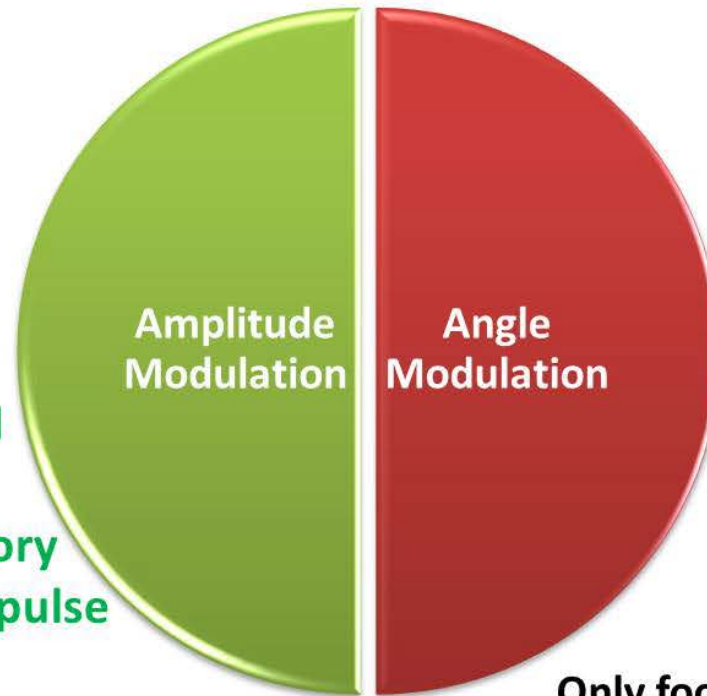
Channel



What is more ...

Amplitude of carrier contains information

- Sinusoid carrier
 - FDM
- Pulse train carrier
- PAM
 - Demod. without ISI
 - Demod. With ISI
 1. Sampling theory
 2. Band-limited pulse
- TDM



Angle of carrier contains information

- Phase modulation
- Frequency modulation
 - Narrowband
 - Wideband

Only focus on some basic properties
The demodulator will be introduced in next semester

Angle modulation

$$c(t) = A \cos(\omega_c t + \theta_c) = A \cos \theta(t)$$

$\theta(t) = \omega_c t + \theta_c$ and where ω_c is the frequency and θ_c the phase of the carrier.

Angle modulation: use modulating signal to change or vary the angle $\theta(t)$

instantaneous frequency $\omega_i(t) = \frac{d\theta(t)}{dt}$ 损失频率

$$y(t) = A \cos[\omega_c t + \theta_c(t)]$$

$$\theta_c(t) = \theta_0 + k_p x(t)$$

Phase modulation: the phase of $y(t)$ is varying with modulating signal

$$y(t) = A \cos \theta(t)$$

$$\frac{d\theta(t)}{dt} = \omega_c + k_f x(t)$$

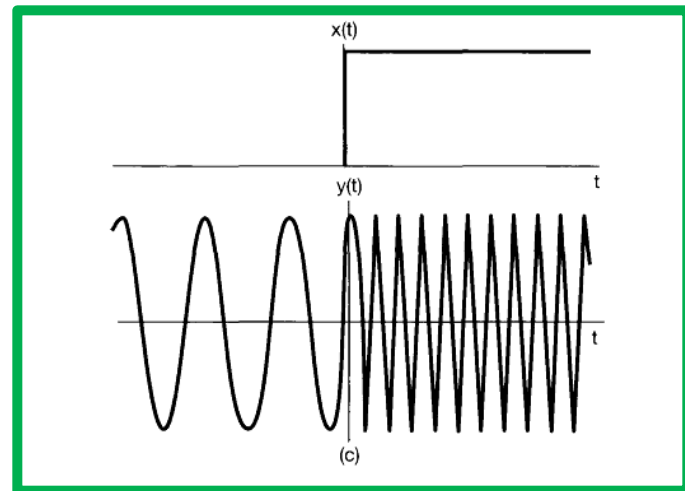
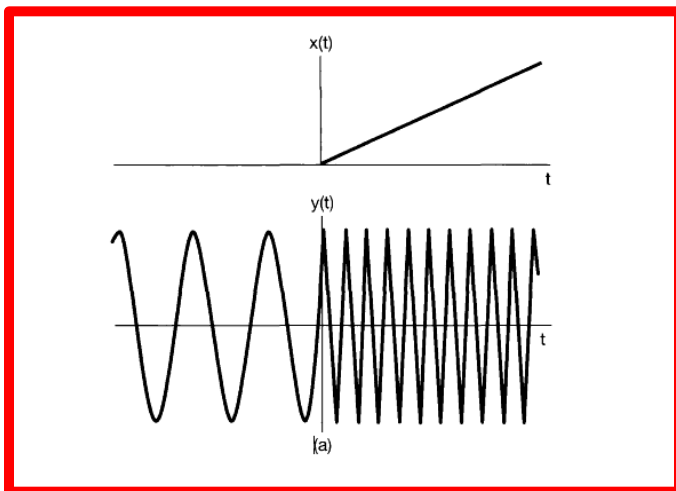
Frequency modulation: the derivative of the angle is varying with modulating signal

$$\frac{d\theta(t)}{dt} = \omega_c + k_p \frac{dx(t)}{dt}$$

$$\frac{d\theta(t)}{dt} = \omega_c + k_f x(t)$$

- Phase modulating with $x(t)$ is identical to frequency modulating with the derivative of $x(t)$.
- Likewise, frequency modulating with $x(t)$ is identical to phase modulating with the integral of $x(t)$.

Fig. 8.32



Narrowband frequency modulation

$$x(t) = A \cos \omega_m t$$

$$\omega_i(t) = \omega_c + k_f A \cos \omega_m t$$

message signal

which varies sinusoidally between $\omega_c + k_f A$ and $\omega_c - k_f A$.

With $\Delta\omega = k_f A$, we have $\omega_i(t) = \omega_c + \Delta\omega \cos \omega_m t$,

$$y(t) = \cos[\omega_c t + \int x(t) dt] = \cos\left(\omega_c t + \frac{\Delta\omega}{\omega_m} \sin \omega_m t + \theta_0\right),$$

where θ_0 is a constant of integration. For convenience we will choose $\theta_0 = 0$, so that

$$y(t) = \cos\left[\omega_c t + \frac{\Delta\omega}{\omega_m} \sin \omega_m t\right].$$

Modulation index for frequency modulation $m = \Delta\omega / \omega_m$

$$\frac{k_f A}{\omega_m}$$

Cont.

- The properties of FM systems tend to be different, depending on whether the modulation index m is small or large.

$$y(t) = \cos(\omega_c t + m \sin \omega_m t)$$

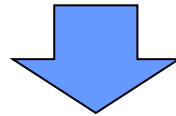
or

$$y(t) = \cos \omega_c t \cos(m \sin \omega_m t) - \sin \omega_c t \sin(m \sin \omega_m t).$$

When m is sufficiently small ($\ll \pi/2$), we can make the approximations

$$\cos(m \sin \omega_m t) \approx 1,$$

$$\sin(m \sin \omega_m t) \approx m \sin \omega_m t,$$



$$y(t) \approx \cos \omega_c t - m(\sin \omega_m t)(\sin \omega_c t).$$

$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t)$$

Similar to AM-DSB/SC

Cont.

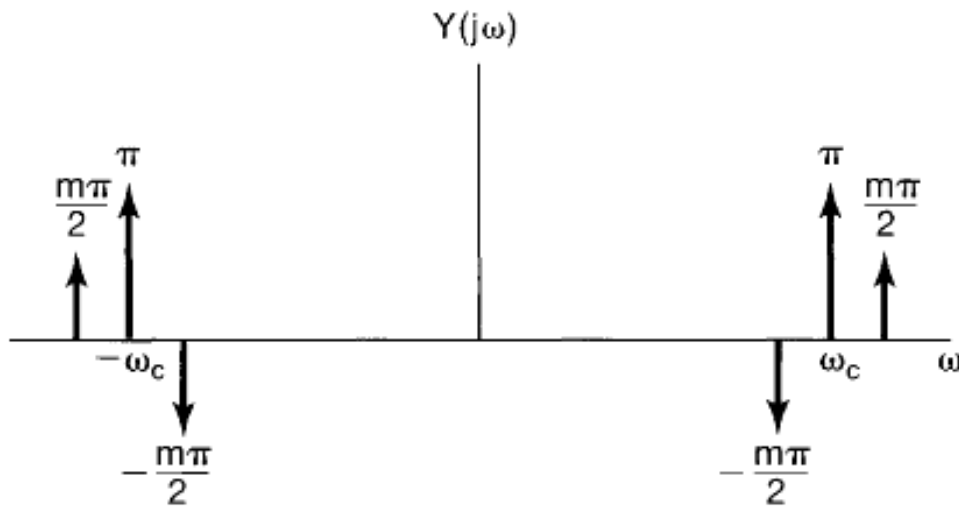


Figure 8.33 Approximate spectrum for narrowband FM.

- The bandwidth of the sidebands is equal to the bandwidth of the modulating signal.
 - ✓ The bandwidth of the sidebands is independent of the modulation index m (i.e., it depends only on the bandwidth of the modulating signal, not on its amplitude).

Wideband Frequency Modulation

$$y(t) = \cos \omega_c t \cos(m \sin \omega_m t) - \sin \omega_c t \sin(m \sin \omega_m t).$$

- The terms $\cos[m \sin \omega_m t]$ and $\sin[m \sin \omega_m t]$ are periodic signals with fundamental frequency ω_m .
- Thus, the Fourier transform of each of these signals is an impulse train with impulses at integer multiples of ω_m and amplitudes proportional to the Fourier series coefficients.
- The coefficients for these two periodic signals involve a class of functions referred to as **Bessel functions** of the first kind.
 - ◆ The first term corresponds to a sinusoidal carrier of the form $\cos \omega_c t$ amplitude modulated by the periodic signal $\cos[m \sin \omega_m t]$
 - ◆ The second term to a sinusoidal carrier $\sin \omega_c t$ amplitude modulated by the periodic signal $\sin[m \sin \omega_m t]$

Cont.

bandwidth

$$B \simeq 2k_f A = 2\Delta\omega.$$

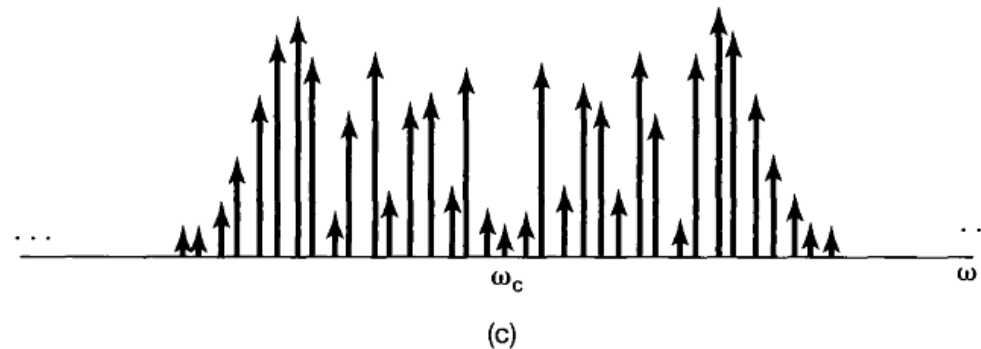
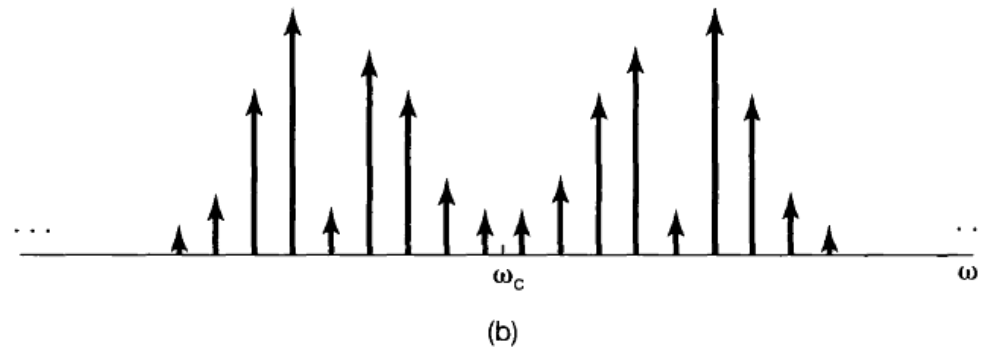
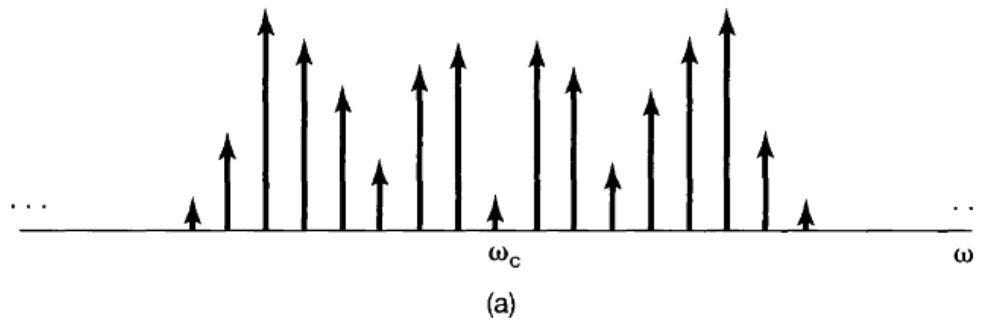


Figure 8.35 Magnitude of spectrum of wideband frequency modulation with $m = 12$: (a) magnitude of spectrum of $\cos \omega_c t \cos[m \sin \omega_m t]$; (b) magnitude of spectrum of $\sin \omega_c t \sin[m \sin \omega_m t]$; (c) combined spectral magnitude of $\cos[\omega_c t + m \sin \omega_m t]$.

Cont.

- For wideband FM, since we assume that m is large, the bandwidth of the modulated signal is much larger than the bandwidth of the modulating signal.
- In contrast to the narrowband case, the bandwidth of the transmitted signal in wideband FM is directly proportional to amplitude A of the modulating signal and the gain factor k_f .